

KINETICS & REACTOR DESIGN

CHE F311 · BITS Pilani · I-Semester 2026-27

Short Notes & Formula Sheet

Mole balances · Sizing · Rate laws · Isothermal design
Rate-data analysis · Multiple reactions · Non-isothermal design
Catalysis · Non-ideal reactors and residence-time distributions

Nine decks, 163 slides

Compiled from: the nine course slide decks in Slides/ — Ch1-krd (1).pdf (12 pp), Ch2-krd (1).pdf (15 pp), Ch3-krd.pdf (16 pp), Ch4-Krd.pdf (23 pp), Ch5-_krd.pdf (19 pp), Ch6_krd.pdf (18 pp), Ch7-_krd.pdf (24 pp), Catalysis_Krd.pdf (14 pp) and Non_ideal_KRD.pdf (22 pp).

Reference text: H. S. Fogler, *Elements of Chemical Reaction Engineering*; O. Levenspiel, *Chemical Reaction Engineering* (the decks' own problem sources).

Companion documents: KRD-Problems-and-Solutions.pdf works every numerical printed in these decks; krd-answers-check.py reproduces every number quoted below and exits 0.

One caveat about a source: Non_ideal_KRD.pdf is headed *Reaction Engineering / CHE G641* — the graduate version of this material. It is the only deck in the set that does not carry the CHE F311 heading, and section 9 says where that shows up.

What is inside

Topic	Deck
Mole balances, rate of reaction, conversion, the four design equations	Ch1
Reactor sizing from conversion; the Levenspiel plot; numerical integration; reactors in series and in parallel; optimum size ratio	Ch2
Rate laws and stoichiometry: molecularity; relative rates; power-law kinetics and the units of k ; the law of mass action; elementary and non-elementary rate laws; reversible laws and K_C ; Arrhenius and activation energy; batch and flow stoichiometric tables; concentration in terms of conversion in liquid and gas systems; the deck's five worked numbers	Ch3
Isothermal reactor design; the algorithm; closed-form conversions; Damköhler number; pressure drop in packed beds	Ch4
Collection and analysis of rate data; differential and integral methods; half-life; differential reactor; nonlinear regression	Ch5
Multiple reactions; selectivity and yield; contacting schemes for the four sign cases; series intermediates; product distribution; net rates	Ch6
Non-isothermal design; energy balance; adiabatic operation; equilibrium thermodynamics; heat exchange; multiple steady states and runaway; interstage cooling and optimum feed temperature	Ch7
Catalysis and catalytic reactors; adsorption and site balance; Langmuir–Hinshelwood; internal and external diffusion	Catalysis
Non-ideal reactors; residence-time distribution; segregation; tanks-in-series; dispersion; two-parameter models	Non-ideal

Notation used throughout

X	conversion of A	τ	space time, V/v_0
$-r_A$	rate of disappearance of A	Da	Damköhler number
F_{A0}, F_A	molar flow of A in / out	k, A, E	rate constant, pre-exponential, activation energy
C_A, C_{A0}	concentration of A / at inlet	ε, δ	volume-change factor, mole-change number
V, W	reactor volume, catalyst mass	K_C, X_e	equilibrium constant, equilibrium conversion
N_A, N_{A0}	moles of A / initially	ΔH_{Rx}	heat of reaction
Θ_i	C_{i0}/C_{A0} , feed ratio	Ua, T_a	heat-transfer coefficient $\times$ area, coolant temperature
p	P/P_0 , reduced pressure	ϕ, η	Thiele modulus, effectiveness factor
α	pressure-drop parameter	$E(t), F(t)$	RTD function, cumulative RTD
t_m, σ^2	mean residence time, variance	Pe, n	Péclet number, tanks-in-series number
$S_{D/U}, \tilde{S}_{D/U}$	selectivity, instantaneous / overall	$\tilde{Y}_D$	yield, rate basis / mole basis

1 Mole Balances, Rates of Reaction and Conversion

Source: Ch1-krd (1).pdf.

1.1 Rate of reaction and the bases it can be written on

- **Rate of reaction** is the number of moles of a species reacting per unit time per unit *extensive quantity* of the system. The extensive quantity is what distinguishes the bases:

$-r_A = \text{mol}/(\text{volume} \cdot \text{time})$	homogeneous: per reactor volume V
$-r'_A = \text{mol}/(\text{mass cat.} \cdot \text{time})$	heterogeneous: per catalyst mass W
$r''_A = \text{mol}/(\text{area} \cdot \text{time})$	per catalyst surface area S
$-r'_A W = r''_A S = -r_A V \quad \Rightarrow \quad \boxed{\text{all three give the same molar rate}}$	

Choosing the wrong basis is the most common error in packed-bed work: the design equation must be written on the same basis as the rate law.

1.2 Conversion, extent and the general mole balance

$X = \frac{\text{moles of A reacted}}{\text{moles of A fed}}$	conversion, defined on the limiting reactant
$N_i = N_i^0 + \nu_i \alpha$	extent of reaction α
$F_{j0} - F_j + G_j = \frac{dN_j}{dt}$	general mole balance on species j

- G_j is the rate of generation of j by reaction within the system volume — $G_j = r_j V$. Everything else in the balance is a transport term.
- Conversion is always defined on the **limiting** reactant, so that $0 \leq X \leq 1$.

1.3 The four design equations

Batch: $\frac{dN_A}{dt} = r_A V$	$\Rightarrow \quad t = N_{A0} \int_0^X \frac{dX}{-r_A V}$
CSTR: $V = \frac{F_{A0} X}{(-r_A)_{\text{exit}}}$	rate evaluated at <i>exit</i> conditions
PFR: $\frac{dF_A}{dV} = r_A$	$\Rightarrow \quad V = F_{A0} \int_0^X \frac{dX}{-r_A}$
PBR: $\frac{dF_A}{dW} = r'_A$	$\Rightarrow \quad W = F_{A0} \int_0^X \frac{dX}{-r'_A}$

- The structural difference between a **CSTR** and a **PFR** is exactly the difference between evaluating $-r_A$ at the exit and integrating it along the path. That single fact drives all of section 2.
- **Space time** $\tau = V/v_0$; **space velocity** $= v_0/V = 1/\tau$.

1.4 Worked from the deck

- **Rocket engine** (slide 36). Chamber 75×60 cm, exhaust 108 kg/s.

$V = 0.2121 \text{ m}^3$	$\Rightarrow$	$-r_{\text{H}_2} = 28 \text{ kmol m}^{-3} \text{ s}^{-1}, \quad -r_{\text{O}_2} = 14 \text{ kmol m}^{-3} \text{ s}^{-1}$
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The 2:1 ratio of the two rates is the stoichiometry of $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ — the first appearance of the relative-rate rule developed in section 3.

- **A→B at three duties** (slide 54). The deck asks for the volume of a CSTR and a PFR, and the batch time, at successively harsher conditions:

	(a)	(b)	(c)
CSTR	99.0 dm ³	2750 dm ³	66 000 dm ³
PFR	99.0 dm ³	127.9 dm ³	660 dm ³
Batch time	9.90 h	12.79 h	66.0 h

In (a) the two reactors are *equal* because the rate is constant there; from (b) onward the CSTR grows far faster than the PFR, which is the Levenspiel-plot argument of the next section in numerical form.

2 Conversion and Reactor Sizing

Source: Ch2-krd (1).pdf.

2.1 The Levenspiel plot

- Plot $F_{A0}/(-r_A)$ **against** X . Then the two design equations become areas on that single graph:

$$V_{\text{CSTR}} = \text{rectangle} = \left. \frac{F_{A0}}{-r_A} \right|_X \cdot X \quad \text{height at the exit} \times \text{width}$$

$$V_{\text{PFR}} = \text{area under the curve} = \int_0^X \frac{F_{A0}}{-r_A} dX \quad \text{integral from 0 to } X$$

- For any *positive-order* kinetics $-r_A$ falls as X rises, so $F_{A0}/(-r_A)$ rises and the **rectangle always exceeds the area under the curve**: a CSTR needs more volume than a PFR for the same conversion. The deck's own numbers in §1.4 show the gap opening from 0 to a factor of 100.

2.2 Numerical integration of the PFR integral

$$\begin{aligned} \text{Simpson (3 pt):} \quad \int &\approx \frac{h}{3} (f_0 + 4f_1 + f_2) \\ \text{Simpson (5 pt):} \quad \int &\approx \frac{h}{3} (f_0 + 4f_1 + 2f_2 + 4f_3 + f_4) \\ \text{3/8 rule:} \quad \int &\approx \frac{3h}{8} (f_0 + 3f_1 + 3f_2 + f_3) \end{aligned}$$

The five-point Simpson rule is the one the deck uses throughout, and it is the one that reproduces the printed answers.

2.3 Reactors in series

$$\begin{aligned} F_{Ai} &= F_{A0} (1 - X_i) && \text{flow leaving stage } i \\ V_i &= \frac{F_{A0} (X_i - X_{i-1})}{(-r_A)_i} && \text{CSTR } i : \text{rectangle from } X_{i-1} \text{ to } X_i \\ V_i &= F_{A0} \int_{X_{i-1}}^{X_i} \frac{dX}{-r_A} && \text{PFR } i : \text{area over the same interval} \end{aligned}$$

2.4 Reactors in parallel

- Rule:** for the total volume to be a minimum the feed must be split so that V/F (**equivalently** τ) **is the same in every branch**. Branches running at different τ waste volume.

2.5 Choosing the best arrangement

- Work from the $1/(-r_A)$ **versus** C_A curve.
- $n > 1$ (rate falls steeply): plug flow first, then a small mixed reactor, then a large mixed reactor.
- $n < 1$: the order is reversed — mixed flow first.
- Two CSTRs, minimum total volume:** maximise the inscribed rectangle, which gives the tangency condition $-dy/dx = y/x$ on the $1/(-r_A)$ curve.

2.6 Worked from the deck — the sizing table (slides 19–38)

The deck gives $T = 500$ K, $P = 830$ kPa, $F_{A0} = 0.4$ mol/s and this rate table. Note that the third row is $-r_A$ itself, not its reciprocal — the reciprocal is the last column.

X	0	0.1	0.2	0.4	0.6	0.7	0.8
$-r_A$ (mol m ⁻³ s ⁻¹)	0.45	0.37	0.30	0.195	0.113	0.079	0.05
$F_{A0}/(-r_A)$ (m ³)	0.89	1.08	1.33	2.05	3.54	5.06	8.00

$$V_{\text{CSTR}} = 8.00 \times 0.8 = 6.40 \text{ m}^3 \quad \text{rectangle to } X = 0.8$$

$$V_{\text{PFR}} = \frac{0.2}{3} (0.89 + 4(1.33) + 2(2.05) + 4(3.54) + 8.00) = \boxed{2.165 \text{ m}^3} \quad \text{5-pt Simpson}$$

$$V_{\text{CSTR}} - V_{\text{PFR}} = 6.40 - 2.165 = 4.235 \text{ m}^3 \quad \text{the price of back-mixing}$$

- **Two CSTRs in series** (slide 9), split at $X = 0.4$:

$$V_1 = 2.05 \times 0.4 = 0.82 \text{ m}^3, \quad V_2 = 8.00 \times (0.8 - 0.4) = 3.20 \text{ m}^3 \Rightarrow \boxed{4.02 \text{ m}^3}$$

- **Two PFRs in series** (slide 10), same split:

$$V_1 = 0.551 \text{ m}^3, \quad V_2 = 1.614 \text{ m}^3 \Rightarrow \boxed{2.165 \text{ m}^3} \quad (\text{identical to one PFR})$$

Two points worth keeping. First, splitting a PFR changes nothing — the integral is additive — whereas splitting a CSTR cuts the volume from 6.40 to 4.02 m³, a 37 % saving. Second, the series-CSTR total still exceeds the PFR, and keeps falling as more stages are added, approaching the PFR area as $N \rightarrow \infty$.

- **Unknown stoichiometry from an MFR** (slide 42). $C_{A0} = 0.10$, $C_{B0} = 0.01$ mol/L, $V = 1$ L, outlet 0.02/0.03/0.04:

$$r_A = -0.08, \quad r_B = +0.02, \quad r_C = +0.04 \Rightarrow \boxed{4A \rightarrow B + 2C}$$

- **Three PFRs in two parallel branches** (slide 46). Branch D holds 50 + 30 L, branch E holds 40 L; equal τ requires

$$\text{fraction to D} = \frac{50}{90} = \boxed{0.556}$$

- **P2-7** (slide 59), $1/(-r_A) = 1.0, 1.67, 5.0, 5.0, 5.0, 5.0, 1.25, 0.91$ at $X = 0, 0.2, 0.4, 0.45, 0.5, 0.6, 0.8, 0.9$:

- | | |
|-----|--|
| (a) | CSTR 24.0 dm ³ ; PFR 71.9 dm ³ for $X = 0.8$ |
| (b) | the two volumes are equal at $X \approx 0.69$ |
| (c) | a 105 dm ³ CSTR gives $X = 0.70$ |
| (d) | CSTR then PFR: $X_1 = 0.401$, $X_2 = 0.643$ |
| (e) | PFR then CSTR: $X_1 = 0.400$, $X_2 = 0.900$ |

(d) and (e) are the same two reactors in the opposite order and differ by 0.26 in exit conversion — the $1/(-r_A)$ curve here is *non-monotonic*, so the ordering rule of §2.5 actually bites.

3 Rate Laws and Stoichiometry

Source: Ch3-krd.pdf — a two-part deck of 63 printed slides: **Part 1, rate laws** (slides 3–28) and **Part 2, stoichiometry** (slides 29–60), closing with three numericals (slides 61–63). Slide numbers cited below are the deck's own printed numbers, read off the slide photographs together with the handwritten callouts.

3.1 Molecularity — how many particles meet (slide 4)

- **Molecularity** is the number of atoms, ions or molecules involved in a single reaction event. The deck classifies reactions by this count and gives one example of each:

unimolecular:	$^{238}\text{U} \longrightarrow ^{234}\text{Th} + ^4\text{He}$	(a single nucleus disintegrates)
bimolecular:	$\text{Br}^\cdot + \text{C}_2\text{H}_6 \longrightarrow \text{HBr} + \text{C}_2\text{H}_5^\cdot$	(a radical meets a molecule)
trimolecular:	$2\text{NO} + \text{Cl}_2 \longrightarrow 2\text{NOCl}$	(three particles in one event)

- **Termolecular events are already rare, and molecularity 4 is the wall:** the deck states in red that *there is no known reaction with molecularity 4* — four particles cannot arrive with the right energy and geometry at the same instant.
- **Why the first example is on this slide at all:** the uranium decay shows that “reaction” in kinetics is not limited to molecules in a beaker; the counting rules of this chapter apply to any transformation written $aA + bB \rightarrow cC + dD$.
- **Net reactions hide the real molecularity** (slide 4): most reaction pathways are a *series of bimolecular steps*. The deck’s illustration is the overall trimolecular-looking $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$, which in fact proceeds through bimolecular encounters. Molecularity belongs to an *elementary step*; overall order (section 3.3) belongs to the *measured* rate law, and the two need not agree — the theme of slides 14–17.

3.2 Relative rates of reaction (slides 5–7)

- Every species in $aA + bB \rightarrow cC + dD$ has its own rate of reaction r_i , but the stoichiometry locks the ratios. Divide each species’ rate by its own stoichiometric coefficient *taken with sign* (negative for reactants, positive for products); all four quotients are equal (slide 5, callout “Reaction Stoichiometry”):



$$\frac{r_A}{-a} = \frac{r_B}{-b} = \frac{r_C}{c} = \frac{r_D}{d} \quad \Longrightarrow \quad r_B = \frac{b}{a} r_A$$

- **Sign convention, stated once and used everywhere:** $r_i > 0$ means species i is being *formed*, $r_i < 0$ means it is being *consumed*. The rate of *disappearance* of a reactant is $-r_i > 0$. Mixing the two readings is the classic way to lose a sign.
- **Why this closes Chapter 1:** the rocket-engine problem of section 1.4 balances $-r_{\text{H}_2}$ against $-r_{\text{O}_2}$ exactly this way (2:1, the coefficients of $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$).

3.2b The two worked relative-rate problems (slides 5–7)

Example (slides 5 and the sticky note). For $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ the rule reads

$$\frac{r_{\text{NO}}}{-2} = \frac{r_{\text{O}_2}}{-1} = \frac{r_{\text{NO}_2}}{2}.$$

The handwritten note asks: “If NO_2 formed at $4 \text{ mol/m}^3\cdot\text{s}$, ($r_{\text{NO}_2} = +4 \text{ mol/m}^3\cdot\text{s}$), what is the rate of formation of NO ?”

$$r_{\text{NO}} = \frac{-2}{2} (+4) = -4 \text{ mol m}^{-3}\text{s}^{-1} \text{ and } r_{\text{O}_2} = \frac{-1}{2} (+4) = -2 \text{ mol m}^{-3}\text{s}^{-1}.$$

The *rate of formation* of NO is $-4 \text{ mol m}^{-3}\text{s}^{-1}$: the minus sign is the answer — NO is not formed at all, it is consumed at $4 \text{ mol m}^{-3}\text{s}^{-1}$.

Exercise (slide 6), solved on slide 7. The reaction $2A + 3B \rightarrow 5C$ is carried out in a reactor. At a particular point the rate of disappearance of A is $10 \text{ mol/dm}^3\cdot\text{s}$. What are the rates of B and C ?

$$\begin{aligned} \frac{r_A}{-2} = \frac{r_B}{-3} = \frac{r_C}{5}, & \quad \text{given } -r_A = 10 \text{ mol/dm}^3/\text{s, i.e. } r_A = -10; \\ r_B = \frac{-3}{-2} (-10) = -15 \text{ mol dm}^{-3}\text{s}^{-1}, & \quad r_C = \frac{5}{-2} (-10) = +25 \text{ mol dm}^{-3}\text{s}^{-1}. \end{aligned}$$

Read the result physically: B is a reactant, so $r_B < 0$ (it disappears at $15 \text{ mol/dm}^3\cdot\text{s}$); C is a product, so $r_C > 0$ (it appears at 25). The three rates are in the ratio 2:3:5 — the stoichiometry again.

3.3 The rate law: what it is and how it is built (slides 8, 12–13)

- **Definition (slide 12, callout bubble):** a *rate law* is a kinetic expression that gives the relationship between reaction rate and concentration. The rate of disappearance depends on *temperature* and *composition*, and the law is built by splitting those two dependencies:

$$-r_A = \underbrace{[k_A(T)]}_{\text{temperature}} \underbrace{[f_n(C_A, C_B, \dots)]}_{\text{composition}}$$

- The rate constant k_A carries *all* the temperature dependence (section 3.7); everything about *who* reacts with *whom* sits in the composition function.
- The subscript A in k_A exists because the numerical value of k depends on *which species' rate* is being written. When the stoichiometric coefficient of every species is 1 — the deck's example $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$ (slide 13) — the subscripts may be dropped:

$$k = k_{\text{NaOH}} = k_{\text{HCl}} = k_{\text{NaCl}} = k_{\text{H}_2\text{O}}.$$

3.4 Power-law kinetics, orders and the units of k (slides 8–10)

- The working assumption of this chapter (slide 8): the reaction $\sum_j \nu_j A_j = 0$ takes place in a **closed, isothermal, constant-pressure** system — a batch at uniform T and P . Then the composition function is a *power law*, a product of concentrations raised to exponents:

$$\begin{aligned} r &= k C_1^{q_1} C_2^{q_2} \dots C_N^{q_N} = k \prod_{j=1}^N C_j^{q_j} && \text{(general power law, slide 8)} \\ -r_A &= k C_A^\alpha C_B^\beta && \text{(two-reactant form)} \\ n &= \alpha + \beta && \text{overall order = sum of the partial orders} \end{aligned}$$

- **Language of orders (slide 8):** the reaction is α order with respect to reactant A and β order with respect to reactant B ; $n = \alpha + \beta$ is the *overall order*. Orders measure how violently the rate responds to each concentration: double C_A and $-r_A$ multiplies by 2^α .
- **Law of mass action (slide 10):** the exponents are *not* free parameters at this stage. For the extent-based rate r of $\sum_j \nu_j A_j = 0$ the deck writes q_j in terms of the stoichiometric coefficients ν_j (negative for reactants, positive for products):

$$r = k \prod_{j=1}^N C_j^{q_j}, \quad q_j = \frac{1}{2}(|\nu_j| - \nu_j)$$

- The combination $\frac{1}{2}(|\nu_j| - \nu_j)$ is a neat switch: it returns $|\nu_j|$ for a reactant ($\nu_j < 0$) and 0 for a product ($\nu_j > 0$) — so the forward power law involves reactant concentrations only. Section 3.5 needs the mirror combination $\frac{1}{2}(|\nu_j| + \nu_j)$, which does the opposite (products only).
- **The units of k betray the order (slide 9).** Write any rate law as $-r_A = k C_A^n$ and demand that both sides carry (concentration)/(time):

$$k = \frac{(\text{Concentration})^{1-n}}{\text{Time}}$$

Order	Rate law ($A \rightarrow$ products)	Units of k (deck's print)
Zero ($n = 0$)	$-r_A = k_A$	$\{k\} = \text{mol dm}^{-3} \text{s}^{-1}$
First ($n = 1$)	$-r_A = k_A C_A$	$\{k\} = \text{s}^{-1}$
Second ($n = 2$)	$-r_A = k_A C_A^2$	$\{k\} = \text{dm}^3 \text{mol}^{-1} \text{s}^{-1}$
Third ($n = 3$)	$-r_A = k_A C_A^3$	$\{k\} = (\text{dm}^3/\text{mol})^2 \text{s}^{-1}$

Practical use: convert units of concentration first (mol/L , mol/m^3 , mmol/L ...), because k changes its number whenever the concentration unit changes. A rate constant quoted in s^{-1} is first order *whatever* the chemistry claims; one quoted in $(\text{dm}^3/\text{mol})^2 \text{s}^{-1}$ is third order.

3.5 Elementary, non-elementary and elementary-looking laws (slides 14–17)

- **Elementary reaction (slide 14):** a chemical reaction in which one or more of the chemical species react directly to form products in a *single reaction step* and with a *single transition state*. Prototypes: $A \rightarrow$ products (unimolecular) and $A + B \rightarrow$ products (bimolecular).
- **Elementary rate law (slide 14):** the reaction *follows the elementary rate law* if the stoichiometric coefficients are **identical** to the reaction order of each species. Then the law of mass action (section 3.4) is literally true and no experiment can disagree.

- **The trap (slide 14):** $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$ is flagged on the slide as a *non-elementary* reaction — its mechanism is a series of bimolecular steps — *but* its measured law $-r_{\text{NO}} = k_{\text{NO}} C_{\text{NO}}^2 C_{\text{O}_2}$ happens to look exactly like the elementary form (orders 2 and 1 = coefficients). Matching coefficients is therefore *not* proof of an elementary mechanism; it is only permission to use the simple law.
- **Non-elementary rate laws (slides 15–17)** are those that *do not follow* simple power laws — fractional orders, denominators, products in numerators. The deck’s catalogue:

(a) **Homogeneous, fractional orders (slide 15).** Gas-phase synthesis of phosgene, $\text{CO} + \text{Cl}_2 \rightarrow \text{COCl}_2$:

$$-r_{\text{CO}} = k C_{\text{CO}} C_{\text{Cl}_2}^{3/2} \quad \text{first order in CO, three-halves in Cl}_2, \text{ five-halves overall.}$$

Half-integer orders cannot come from any single elementary step — they are the fingerprint of a hidden mechanism.

(b) **Heterogeneous, on a catalyst (slides 16–17; the deck prints this example on both).** Gas-solid catalysed *hydrodemethylation of toluene (T)*: $\text{C}_6\text{H}_5\text{CH}_3 + \text{H}_2 \xrightarrow{\text{cat}} \text{C}_6\text{H}_6 + \text{CH}_4$, written $T + \text{H}_2 \rightarrow B + M$. The rate of disappearance of toluene *per mass of catalyst* (hence the prime on r'):

$$-r'_T = \frac{k P_H P_T}{1 + K_B P_B + K_T P_T} \quad K_B, K_T = \text{adsorption constants.}$$

Two things to notice (slide 16 callout): the law is written *in terms of partial pressures rather than concentrations* — on a surface the coverage, not the molarity, controls the rate — and the denominator is the adsorption competition of products and reactants for the same sites (developed fully in section 8). Partial pressures and concentrations are interchangeable through $P_i = C_i RT$ only in ideal gases; the *form* of the law is what matters here.

(c) **Cumene dehydrogenation (slide 24, listing C).** Cumene (C) $\rightarrow$ benzene (B) + propylene (P) over a catalyst, again per mass of catalyst:

$$-r_C = \frac{k \left(P_C - \frac{P_B P_P}{K} \right)}{1 + K_B P_B + K_C P_C}$$

The numerator is *forward – reverse* (the reaction is reversible; section 3.6); the denominator again collects adsorption terms. This single line already combines every ingredient of this chapter and of section 8.

3.6 Reversible rate laws (slides 11, 18–21)

The thermodynamic straitjacket (slide 18). For $aA + bB \rightleftharpoons cC + dD$, *all rate laws must reduce to the thermodynamic relationship relating the reacting species concentrations at equilibrium* — the slide’s orange callout names it the Thermodynamic Equilibrium Relationship:

$$K_C = \frac{C_{Ce}^c C_{De}^d}{C_{Ae}^a C_{Be}^b} \quad (\text{concentrations at equilibrium carry the subscript } e)$$

At equilibrium the net rate must vanish; therefore a reversible rate law is always a *difference of two terms*, forward minus reverse, and the ratio of the two rate constants is tied to K_C :

$$r = k_f \prod_{j=1}^N C_j^{q_j} - k_r \prod_{j=1}^N C_j^{q_j^*} \quad (\text{slide 11, general form})$$

$$q_j = \frac{1}{2}(|\nu_j| - \nu_j), \quad q_j^* = \frac{1}{2}(|\nu_j| + \nu_j)$$

The pair $\frac{1}{2}(|\nu_j| \mp \nu_j)$ splits every species into “reactant side” (q_j) or “product side” (q_j^*), so the forward product runs on reactant concentrations and the reverse product on product concentrations. Setting $r = 0$ and rearranging gives $k_f/k_r = K_C$ — exactly the thermodynamic constraint above.

Worked in full: benzene dimerisation (slides 19–21). The deck’s reaction is $2\text{C}_6\text{H}_6 \rightleftharpoons \text{C}_{12}\text{H}_{10} + \text{H}_2$, written symbolically $2B \rightleftharpoons D + H_2$ (diphenyl D). Build the law step by step as the slides do:

Step 1 — the two one-way rates (slide 19). Forward: benzene disappears, second order; reverse: diphenyl and hydrogen recombine, first order in each.

$$-r_{B,\text{forward}} = k_B C_B^2 \quad (\text{equivalently } r_{B,\text{forward}} = -k_B C_B^2)$$

$$r_{B,\text{reverse}} = k_{-B} C_D C_{H_2}$$

Step 2 — net formation of benzene (slide 20):

$$r_B \equiv r_{B,\text{net}} = r_{B,\text{forward}} + r_{B,\text{reverse}} = -k_B C_B^2 + k_{-B} C_D C_{H_2}$$

Step 3 — multiply by -1 to get the rate of *disappearance* of benzene, and factorise the forward constant out:

$$-r_B = k_B C_B^2 - k_{-B} C_D C_{H_2} = k_B \left(C_B^2 - \frac{k_{-B}}{k_B} C_D C_{H_2} \right)$$

Step 4 — replace the constant ratio by K_C (slide 21): since $k_B/k_{-B} = K_C$ (Concentration equilibrium constant),

$$-r_B = k_B \left(C_B^2 - \frac{C_D C_{H_2}}{K_C} \right)$$

The structure of the boxed law is the reusable part: (*forward term*) $\times (1 - (\text{reverse})/(\text{forward without } k))$. At equilibrium the bracket vanishes, recovering $K_C = C_D C_{H_2}/C_B^2$ — which is slide 18's thermodynamic check done on a real example. Large K_C pushes the bracket toward 1 and the reaction behaves irreversibly; chapter 5 uses exactly this limiting behaviour.

3.7 A catalogue of rate laws from the deck (slides 22–24)

Slides 22–24 print Fogler's catalogue of archetypal rate laws. Learn the *shapes*; each one reappears in section 5 (how the shape is found from data) and section 8 (why the shape has a denominator).

- **A. First-order laws (slide 22)** — each law is first order *overall*; one reversible entry:

No.	Reaction	Rate law as printed
1	$C_2H_6 \rightarrow C_2H_4 + H_2$ (alkane dehydrogenation)	$-r_A = k C_{C_2H_6}$
2	$\varphi N=NCl \rightarrow \varphi Cl + N_2$ (diazo chloride decomposes; φ = phenyl)	$-r_A = k C_{\varphi N=NCl}$
3	ethylene oxide + $H_2O \xrightarrow{H_2SO_4}$ ethylene glycol	$-r_A = k C_{CH_2OCH_2}$ (the printed epoxide ring)
4	$CH_3COCH_3 \rightarrow CH_2CO + CH_4$ (acetone $\rightarrow$ ketene)	$-r_A = k C_{CH_3COCH_3}$
5	$nC_4H_{10} \rightleftharpoons iC_4H_{10}$ (isomerisation, <i>reversible</i>)	$-r_A = k(C_{nC_4H_{10}} - C_{iC_4H_{10}}/K_C)$

- **B. Second-order laws (slide 23)** — two irreversible, one reversible (the ester interchange $A + B \rightleftharpoons C + D$):

No.	Reaction	Rate law as printed
1	$p\text{-nitrochlorobenzene} + 2NH_3 \rightarrow p\text{-nitroaniline} + NH_4Cl$	$-r_A = k C_{ClC_6H_4NO_2} C_{NH_3}$
2	$CNBr + CH_3NH_2 \rightarrow CH_3Br + NCNH_2$	$-r_A = k C_{CNBr} C_{CH_3NH_2}$
3	$CH_3COOC_2H_5 + C_3H_7OH \rightleftharpoons CH_3COOC_3H_7 + C_2H_5OH$	$-r_A = k[C_A C_B - C_C C_D/K_C]$

- **C. Non-elementary laws (slide 24).** (1) Acetaldehyde cracking, $CH_3CHO \rightarrow CH_4 + CO$, is *three-halves* order: $-r_{CH_3CHO} = k C_{CH_3CHO}^{3/2}$. (2) The cumene law of section 3.5(c).

- **D. Enzymatic reaction (slide 24).** Urea (U) hydrolysed by the enzyme urease (E), $NH_2CONH_2 + H_2O \xrightarrow{\text{urease}} 2NH_3 + CO_2$ — the Michaelis–Menten shape:

$$-r_U = \frac{k_3 C_E C_U}{K_M + C_U} \quad (\text{saturates at high } C_U: \text{ the enzyme runs out of free time})$$

- **E. Biomass reaction (slide 24).** Substrate (S) + Cells (C) $\rightarrow$ More cells + product — living catalysts give the same saturating shape with cells in the numerator:

$$-r_S = \frac{k_3 C_C C_S}{K_M + C_S}$$

- **Footnote on the slide:** the rate constants and activation energies for a number of these reactions are given in the Data Bank on the CD-ROM and Summary Notes of the reference text — the catalogue teaches *forms*, not numbers.

3.8 The rate constant $k(T)$: Arrhenius (slides 25–28)

- **The Arrhenius equation (slide 25, callout bubble):** all the temperature dependence of a rate law hides inside $k_A(T)$:

$$k_A(T) = A e^{-E/RT}$$

A = pre-exponential factor (frequency factor)

E = activation energy, J/mol or cal/mol

R = gas constant = $8.314 \text{ J mol}^{-1} \text{ K}^{-1} = 1.987 \text{ cal mol}^{-1} \text{ K}^{-1}$

T = absolute temperature, K

- **Limits (the k -vs- T curve on slide 25):** $T \rightarrow \infty$ gives $k \rightarrow A$ — the pre-exponential factor is the ceiling of the rate constant — and $T \rightarrow 0$ gives $k \rightarrow 0$. The curve is flat at low T , steep in the middle, and levels off at A ; almost every industrial reactor lives in the steep middle.
- **Collision reading of the two factors (slides 25–26):**

A : total number of collisions

$e^{-E/RT}$: probability that a collision results in a reaction

k : number of collisions that result in a reaction

- **Activation energy (slide 26)** is a measure of the *minimum energy the reacting molecules must have* in order for the reaction to occur — the energy required to reach the transition state. The slide's energy diagram: reactants $AB + C$ climb over the barrier to the activated complex $A \cdots B \cdots C$ (transition state at the saddle) and roll down to products $A + BC$. The barrier height is E ; a catalyst works by lowering it (section 8), never by changing the end states.
- **Logarithmic form (slide 27)** — the working form for data analysis:

$$k_A(T) = A e^{-E/RT} \quad \implies \quad \ln k_A = \ln A - \frac{E}{R} \cdot \frac{1}{T}$$

- **The slide's E - k - r shorthand (slide 27):** at a given temperature the activation energy and the rate constant move in *opposite* directions ($E \uparrow \Rightarrow k \downarrow \Rightarrow r \downarrow$; $E \downarrow \Rightarrow k \uparrow \Rightarrow r \uparrow$). Everything the temperature does to a reactor passes through this one exponential.
- **Two-temperature form** — subtract the logarithmic form written at two temperatures (the form the deck itself uses in the slide-61 pasteurisation problem):

$$\boxed{\ln \frac{k_2}{k_1} = \frac{E}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)} \quad \iff \quad E = \frac{R \ln(t_1/t_2)}{1/T_1 - 1/T_2} \quad (\text{equal first-order conversions})$$

- **The Arrhenius plot (slide 28)** is $\ln k$ vs $1/T$: a straight line of slope $-E/R$ and intercept $\ln A$. Two runs give E without knowing A ; five runs give a regression (section 5). The slide's conclusions, verbatim in substance:
 1. Reactions with high activation energies are very temperature-sensitive; reactions with low activation energies are relatively temperature-insensitive.
 2. Any given reaction is much more temperature-sensitive at a low temperature than at a high temperature.
 (Callout bubble:) *the larger the activation energy, the more temperature sensitive k and thus the reaction rate.*

Point 2 is quantitative: $\frac{d \ln k}{dT} = \frac{E}{RT^2}$ — the same E produces a far bigger fractional change in k at small T . This single derivative reappears as the slope of van't Hoff-style plots and as the sensitivity knob in the non-isothermal design of section 7.

3.9 Part 2 opens: why stoichiometric tables (slides 29–32)

- **Purpose of developing a stoichiometric table (slide 29):** to determine the number of moles of each species *remaining* at a conversion X . The rate law needs concentrations; the reactor design equations of Chapter 2 need concentrations as functions of X ; the table is the machine that manufactures them.
- **Components of the table (slide 30):** columns *Species* — *Initially* — *Change* — *Remaining*. The slide's handwritten callout: “*Change refers to moles of species reacted or formed*” — negative entries for what is consumed, positive for what is made. Rows for every reactant and product, one for inerts (I), one for totals.
- **Conversion recalls Chapter 2 (slide 31)** for $aA + bB \rightarrow cC + dD$:

$$X = \frac{N_{A0} - N_A}{N_{A0}} \quad \Rightarrow \quad N_A = N_{A0} - N_{A0}X \quad \Rightarrow \quad \boxed{N_A = N_{A0}(1 - X)}$$

Callouts on slide 31: the numerator $N_{A0} - N_A$ is the *moles of A reacted*; the boxed N_A is the *moles of A remaining in the reactor at a conversion of X*. Everything else is bookkeeping in the same style.

- **Every other row follows from one ratio (slide 32):** moles reacted or formed are tied to moles of A reacted by the stoichiometry,

$$\begin{aligned} \text{moles B reacted} &= \frac{\text{moles B reacted}}{\text{moles A reacted}} \{ \text{moles A reacted} \} = \frac{b}{a} (N_{A0}X) \\ \text{moles C formed} &= \frac{c}{a} (N_{A0}X), \quad \text{moles D formed} = \frac{d}{a} (N_{A0}X) \end{aligned}$$

- Subtract the reacted amount from the initial amount (reactants) or add the formed amount (products) and the remaining moles follow (slide 33):

$$N_B = N_{B0} - \frac{b}{a} N_{A0}X, \quad N_C = N_{C0} + \frac{c}{a} N_{A0}X, \quad N_D = N_{D0} + \frac{d}{a} N_{A0}X$$

3.10 The batch stoichiometric table and δ (slides 34–36)

Species	Initially (mol)	Change (mol)	Remaining (mol)
A	N_{A0}	$-(N_{A0}X)$	$N_A = N_{A0} - N_{A0}X = N_{A0}(1 - X)$
B	N_{B0}	$-\frac{b}{a}(N_{A0}X)$	$N_B = N_{B0} - \frac{b}{a}N_{A0}X$
C	N_{C0}	$+\frac{c}{a}(N_{A0}X)$	$N_C = N_{C0} + \frac{c}{a}N_{A0}X$
D	N_{D0}	$+\frac{d}{a}(N_{A0}X)$	$N_D = N_{D0} + \frac{d}{a}N_{A0}X$
I	N_{I0}	—	$N_I = N_{I0}$
Total	N_{T0}	$\delta N_{A0}X$	$N_T = N_{T0} + \left(\frac{d}{a} + \frac{c}{a} - \frac{b}{a} - 1 \right) N_{A0}X$

- **The total row defines δ (slides 34–35):** add up the changes and the coefficient of $N_{A0}X$ is the change in the *total* number of moles *per mole of A reacted* — the slide's red circle and the phrase “change in the total number of moles per mole of A reacted”:

$$N_T = N_{T0} + \delta N_{A0}X, \quad \boxed{\delta = \frac{d}{a} + \frac{c}{a} - \frac{b}{a} - 1}$$

- **Interpretation:** $\delta > 0$ (moles created, e.g. $A \rightarrow 3R$) makes a constant- T, P gas *expand*; $\delta < 0$ (moles destroyed, e.g. $2A \rightarrow R$) makes it *contract*; $\delta = 0$ (equal moles each side) freezes the total. Inerts do not appear in δ — they never react.
- **Why the table must also carry concentrations (slide 36):** the rate laws waiting downstream are $-r_A = kC_A, kC_A^2, kC_A^3$ — functions of *concentration*, not of moles. The next slides divide every row by the volume.

3.11 Concentration in terms of conversion — batch liquids (slides 36–41)

For a liquid-phase batch the volume is constant, $V = V_0$ — the deck states this explicitly on every liquid slide. Divide the remaining-moles rows by V (slide 37):

$$C_A = \frac{N_A}{V} = \frac{N_{A0}(1-X)}{V}, \quad C_B = \frac{N_B}{V} = \frac{N_{B0} - (b/a)N_{A0}X}{V},$$

$$C_C = \frac{N_C}{V} = \frac{N_{C0} + (c/a)N_{A0}X}{V}, \quad C_D = \frac{N_D}{V} = \frac{N_{D0} + (d/a)N_{A0}X}{V}$$

- **Introduce the feed ratio Θ_i (slides 38–39):** factor N_{A0} out of each bracket and the ratio N_{i0}/N_{A0} appears. The deck's yellow box gives its three equivalent guises (mole, concentration and mole-fraction form):

$$\Theta_i = \frac{N_{i0}}{N_{A0}} = \frac{C_{i0}}{C_{A0}} = \frac{y_{i0}}{y_{A0}}$$

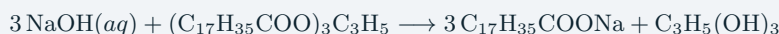
- With $\Theta_B = N_{B0}/N_{A0}$ the product $N_{A0}\Theta_B$ simply is N_{B0} (slide 40 annotation), and every concentration becomes $C_{A0} \times$ (a function of X) — which is the form the rate laws want. The complete liquid-batch set (slides 40–41):

Species	Remaining (mol)	Concentration ($V = V_0$)
A	$N_{A0}(1-X)$	$C_A = \frac{N_{A0}(1-X)}{V} = C_{A0}(1-X)$
B	$N_{B0} - \frac{b}{a}N_{A0}X = N_{A0}\left(\Theta_B - \frac{b}{a}X\right)$	$C_B = C_{A0}\left(\Theta_B - \frac{b}{a}X\right)$
C	$N_{C0} + \frac{c}{a}N_{A0}X = N_{A0}\left(\Theta_C + \frac{c}{a}X\right)$	$C_C = C_{A0}\left(\Theta_C + \frac{c}{a}X\right)$
D	$N_{D0} + \frac{d}{a}N_{A0}X = N_{A0}\left(\Theta_D + \frac{d}{a}X\right)$	$C_D = C_{A0}\left(\Theta_D + \frac{d}{a}X\right)$
I	N_{I0}	$C_I = C_{I0}$

Sign check before use: for a reactant the bracket $\Theta_B - \frac{b}{a}X$ must stay positive up to the maximum physically possible conversion (both species run out together at $X_{\max} = a\Theta_B/b$). A negative bracket always means the assumed X is impossible — the stoichiometry, not the algebra, has failed.

3.12 Worked from the deck: the saponification table (slides 42–44)

Statement (slide 42). “Given the saponification for the formation of soap from aqueous caustic soda & glyceryl stearate is:”



“Letting X the conversion of sodium hydroxide, set up a stoichiometric table expressing the concentration of each species in terms of its initial concentration and the conversion.”

Step 1 — letter the species (slide 43). $3A + B \rightarrow 3C + D$ with $a = 3$, $b = 1$, $c = 3$, $d = 1$ ($A = \text{NaOH}$, $B =$ glyceryl stearate, $C =$ soap, $D =$ glycerol).

Step 2 — liquid phase, so $V = V_0$ and

$$C_A = \frac{N_{A0}(1-X)}{V} = \frac{N_{A0}(1-X)}{V_0} = C_{A0}(1-X), \quad C_B = \frac{N_{A0}[\Theta_B - (b/a)X]}{V_0} = C_{A0}\left[\Theta_B - \frac{1}{3}X\right].$$

Step 3 — fill the table (slide 44); note $(c/a) = 1$ and $(d/a) = \frac{1}{3}$:

Species	Initially	Change	Remaining	Concentration
A	N_{A0}	$-(N_{A0}X)$	$N_A = N_{A0}(1-X)$	$C_A = C_{A0}(1-X)$
B	N_{B0}	$-\frac{1}{3}(N_{A0}X)$	$N_B = N_{A0}(\Theta_B - \frac{1}{3}X)$	$C_B = C_{A0}(\Theta_B - \frac{1}{3}X)$
C	N_{C0}	$+(N_{A0}X)$	$N_C = N_{A0}(\Theta_C + X)$	$C_C = C_{A0}(\Theta_C + X)$
D	N_{D0}	$+\frac{1}{3}(N_{A0}X)$	$N_D = N_{A0}(\Theta_D + \frac{1}{3}X)$	$C_D = C_{A0}(\Theta_D + \frac{1}{3}X)$
I	N_{I0}	—	$N_I = N_{I0}$	$C_I = C_{I0}$
Total	N_{T0}	0	$N_T = N_{T0}$	$(\delta = \frac{1}{3} + 1 - \frac{1}{3} - 1 = 0)$

The total row is the quiet punchline: four moles in, four moles out ($3+1 \rightarrow 3+1$), so $\delta = 0$ and the total moles never move. If this mixture were a gas at constant T and P its volume would be frozen at V_0 — which is why the liquid assumption $V = V_0$ (slide 43) is harmless here even in principle. The companion problem document works this table as Q11.

3.13 The flow-system stoichiometric table (slides 45–48)

- **Purpose (slide 45):** to determine the *effluent flow rate* of each species at a conversion X . The molar flow F_i (mol/time) replaces the moles N_i ; the box on the slide shows feed ($F_{A0}, F_{B0}, F_{C0}, F_{D0}, F_{I0}$) entering a flow reactor that runs $A + \frac{b}{a}B \rightarrow \frac{c}{a}C + \frac{d}{a}D$ and the effluent ($F_A, \dots, F_I$) leaving.

- **Components (slide 46):** columns *Species* — *Feed rate to reactor (mol/time)* — *Change within the reactor (mol/time)* — *Effluent rate from reactor (mol/time)* — and, added on slide 47, *Concentration (mol/L)*.
- **The full table (slide 47):** the batch table with $N \rightarrow F$ throughout, and concentrations from the volumetric flow rate at the relevant point, $C_i = F_i/v$:

Species	Feed rate	Change	Effluent rate	Concentration
A	F_{A0}	$-(F_{A0}X)$	$F_A = F_{A0}(1 - X)$	$C_A = \frac{F_{A0}(1 - X)}{v}$
B	$F_{B0} = \Theta_B F_{A0}$	$-\frac{b}{a}(F_{A0}X)$	$F_B = F_{A0}\left(\Theta_B - \frac{b}{a}X\right)$	$C_B = \frac{F_{A0}[\Theta_B - (b/a)X]}{v}$
C	$F_{C0} = \Theta_C F_{A0}$	$+\frac{c}{a}(F_{A0}X)$	$F_C = F_{A0}\left(\Theta_C + \frac{c}{a}X\right)$	$C_C = \frac{F_{A0}[\Theta_C + (c/a)X]}{v}$
D	$F_{D0} = \Theta_D F_{A0}$	$+\frac{d}{a}(F_{A0}X)$	$F_D = F_{A0}\left(\Theta_D + \frac{d}{a}X\right)$	$C_D = \frac{F_{A0}[\Theta_D + (d/a)X]}{v}$
I	$F_{I0} = \Theta_I F_{A0}$	—	$F_I = F_{I0}$	$C_I = \frac{F_{I0}}{v}$
Total	F_{T0}	$\delta F_{A0}X$	$F_T = F_{T0} + \delta F_{A0}X$	$C_T = \frac{F_T}{v}$

- Here Θ_i is defined on flow rates (slide 49's pink box): $\Theta_i = F_{i0}/F_{A0} = C_{i0}/C_{A0} = y_{i0}/y_{A0}$ — same three guises, same numbers as in a batch charge with the same feed composition.
- **The closing example statement (slide 48):** “Given a liquid phase reaction: $A + 2B \rightarrow C + D$. The initial concentration of A and B are 1.8 kmol/m^3 and 6.6 kmol/m^3 respectively. Construct a stoichiometric table for a flow system considering A as the basis of calculation.”

3.14 Worked from the deck: $A + 2B \rightarrow C + D$ in flow (slides 49–52)

Step 1 — coefficients and feed ratios (slide 49). $a = 1, b = 2, c = 1, d = 1$; $C_{A0} = 1.8, C_{B0} = 6.6 \text{ kmol/m}^3$ and $C_{C0} = C_{D0} = 0$ (the slide annotates: “Since C & D are products”). Then

$$\delta = \frac{d}{a} + \frac{c}{a} - \frac{b}{a} - 1 = 1 + 1 - 2 - 1 = -1, \quad \Theta_B = \frac{C_{B0}}{C_{A0}} = \frac{6.6}{1.8} = 3.67, \quad \Theta_C = \Theta_D = 0.$$

Step 2 — fill the table symbolically (slide 50), then with numbers (slide 51):

Species	Feed rate	Change	Effluent rate (with $F_{B0} = 3.67 F_{A0}$)
A	F_{A0}	$-(F_{A0}X)$	$F_A = F_{A0}(1 - X)$
B	$F_{B0} = 3.67 F_{A0}$	$-2(F_{A0}X)$	$F_B = F_{A0}(3.67 - 2X)$
C	$F_{C0} = 0$	$+(F_{A0}X)$	$F_C = F_{A0}X$
D	$F_{D0} = 0$	$+(F_{A0}X)$	$F_D = F_{A0}X$
Total	F_{T0}	$-(F_{A0}X)$	$F_T = F_{T0} - F_{A0}X \quad (\delta = -1)$

- **Into concentrations (slide 52, liquid, $V = V_0$ in batch; $v = v_0$ in flow):** each row divided by C_{A0} times C_{A0} :

$$C_A = C_{A0}(1 - X), \quad C_B = C_{A0}(3.67 - 2X), \quad C_C = C_{A0}X, \quad C_D = C_{A0}X.$$

B is the limiting reagent if X exceeds $3.67/2 = 1.835$ — which it never can: $X \leq 1$ always, and $\Theta_B = 3.67$ means a large excess of B, so the bracket stays positive with margin. The companion document works this construction as Q12.

3.15 Concentration in terms of conversion — the four cases (slides 52–60)

Part 2's finale is a single master formula assembled case by case. Two questions decide everything: *does the volume change?* (liquids: no; gases: yes) and *is it batch (N) or flow (F)?*

Case 1 — liquid, batch (slide 52): $V = V_0$ (boxed on the slide), so division by V_0 gives at once

$$C_A = C_{A0}(1 - X), \quad C_B = C_{A0}\left[\Theta_B - \frac{b}{a}X\right], \quad C_C = C_{A0}\left[\Theta_C + \frac{c}{a}X\right], \quad C_D = C_{A0}\left[\Theta_D + \frac{d}{a}X\right].$$

Case 2 — liquid, flow (slide 53): liquids are incompressible, $v = v_0$ (boxed), so $C_i = F_i/v = F_i/v_0$ and the same four formulas hold with $N \rightarrow F$ already built in:

$$C_A = \frac{F_{A0}(1 - X)}{v} = C_{A0}(1 - X), \quad C_B = C_{A0}\left[\Theta_B - \frac{b}{a}X\right], \quad \dots$$

Case 3 — gas, batch (slides 54–56): now V must come from the equation of state. The deck writes the real-gas law at any time t and at the initial condition:

$$PV = Z N_T R T \quad (1)$$

$$P_0 V_0 = Z_0 N_{T0} R T_0 \quad (2)$$

with the slide's bubble: T in K, P = total pressure in atm (1 atm = 101.3 kPa), Z = compressibility factor, $R = 0.08206 \text{ dm}^3 \text{ atm mol}^{-1} \text{ K}^{-1}$ (the pressure-units gas constant). Divide (1) by (2) and solve for V (slide 55):

$$V = V_0 \left(\frac{P_0}{P} \right) \frac{T}{T_0} \left(\frac{Z}{Z_0} \right) \frac{N_T}{N_{T0}} \quad (3)$$

If Z does not change significantly during the reaction, $Z_0 \approx Z$ (slide 55 annotation) and the compressibility ratio cancels. It remains to convert the mole ratio — the job of ε (slide 56):

$$\frac{N_T}{N_{T0}} = 1 + \delta y_{A0} X = 1 + \varepsilon X,$$

$$\varepsilon = y_{A0} \delta = \left(\frac{d}{a} + \frac{c}{a} - \frac{b}{a} - 1 \right) \frac{N_{A0}}{N_{T0}}$$

$$V = V_0 \left(\frac{P_0}{P} \right) (1 + \varepsilon X) \frac{T}{T_0} \quad (5, \text{ constant } Z)$$

- **The equivalent readings of ε** (slide 56 gives all of them; they are useful checks on any calculation):

$$\varepsilon = \frac{N_T - N_{T0}}{N_{T0} X}$$

and, at complete conversion ($X = 1$, irreversible, $N_T \rightarrow N_{Tf}$):

$$\varepsilon = \frac{N_{Tf} - N_{T0}}{N_{T0}} = \frac{V_{XA=1} - V_{XA=0}}{V_{XA=0}}$$

Words (slide 56): ε is the *fractional change in volume of the system between no conversion and complete conversion of reactant A*. The annotation stresses that $N_T/N_{T0} = 1 + \varepsilon X$ *applies for both batch and flow systems* — it is a statement about the *gas*, not about the vessel.

The three regimes: $\varepsilon = 0$ (no mole change, or so diluted that $y_{A0} \rightarrow 0$) freezes the volume and all formulas collapse to the liquid ones; $\varepsilon > 0$ expands the gas, diluting everything faster than $(1 - X)$; $\varepsilon < 0$ contracts it. Setting $\varepsilon = 0$ is the single most common legitimate simplification — the deck's own gas examples state it explicitly before using it.

Case 4 — gas, flow (slides 57–60): identical logic on molar flow rates and the volumetric flow v . At any point $C_T = F_T/v = P/(ZRT)$ (1), at the entrance $C_{T0} = F_{T0}/v_0 = P_0/(Z_0RT_0)$ (2); dividing (1) by (2) and using the flow-table total row $F_T = F_{T0} + F_{A0}\delta X$ (slide 58 annotation):

$$v = v_0 \frac{F_T}{F_{T0}} \frac{P_0}{P} \frac{T}{T_0} = v_0 \left(1 + \frac{F_{A0}}{F_{T0}} \delta X \right) \frac{P_0}{P} \frac{T}{T_0} = v_0 (1 + \varepsilon X) \frac{P_0}{P} \frac{T}{T_0} \quad (4)$$

(the same $\varepsilon = y_{A0}\delta$; $F_{A0}/F_{T0} = y_{A0}$). Finally substitute v from (4) and $F_j = F_{A0}(\Theta_j + \nu_j X)$ (5) — the slide annotates ν_j as the stoichiometric coefficient of species j with sign (d/a , c/a , $-b/a$, $-a$) — into $C_j = F_j/v$ (slides 59–60):

$$C_j = C_{A0} \frac{\Theta_j + \nu_j X}{1 + \varepsilon X} \cdot \frac{P}{P_0} \cdot \frac{T_0}{T}$$

$$C_A = C_{A0} \frac{1 - X}{1 + \varepsilon X} \cdot \frac{T_0}{T} \cdot \frac{P}{P_0},$$

$$C_B = C_{A0} \frac{\Theta_B - (b/a)X}{1 + \varepsilon X} \cdot \frac{T_0}{T} \cdot \frac{P}{P_0},$$

$$C_C = C_{A0} \frac{\Theta_C + (c/a)X}{1 + \varepsilon X} \cdot \frac{T_0}{T} \cdot \frac{P}{P_0},$$

$$C_D = C_{A0} \frac{\Theta_D + (d/a)X}{1 + \varepsilon X} \cdot \frac{T_0}{T} \cdot \frac{P}{P_0},$$

$$C_I = \frac{C_{A0} \Theta_I}{1 + \varepsilon X} \cdot \frac{T_0}{T} \cdot \frac{P}{P_0}$$

The three ratios are each easy to misplace; the sanity check is limits. At $X = 0$ every formula must return $C_{A0}\Theta_j$; at constant T, P and $\varepsilon = 0$ they must reduce to the liquid forms; and P/P_0 appears *first power up* because compression ($P > P_0$) concentrates the gas. Note $\nu_A = -a/a = -1$ per the slide's sign list, which is how C_A acquires $(1 - X)$. Sections 4 and 7 plug these $C_j(X)$ expressions straight into the design equations.

3.16 The three closing numericals (slides 61–63)

Pasteurisation (slide 61). “Milk is pasteurized if it is heated to 63 °C for 30 min, but if it is heated to 74 °C it only needs 15 s for the same result. Find the activation energy of this sterilization process.”

Sterilisation is first-order killing: the same result means the same kt . Equating $k_1t_1 = k_2t_2$ with Arrhenius k 's and cancelling A gives the two-temperature form of section 3.8:

$$E = \frac{R \ln(t_1/t_2)}{1/T_1 - 1/T_2} = \frac{8.314 \ln(1800/15)}{1/336.15 - 1/347.15} = \frac{8.314 \times 4.7875}{9.43 \times 10^{-5}} \implies \boxed{E \approx 422 \text{ kJ/mol}}$$

(Two k 's, two unknowns E and A — but A cancels, which is why a two-point sterilisation schedule determines E outright.)

A note on reading the slide: an earlier pass at this photograph misread the 63 °C as 80 °C and concluded the printed pairing must be reversed. The photograph prints 63 °C for 30 min and 74 °C for 15 s — a consistent hot-brief/cool-long pair — and the calculation above follows the print. The enormous E (422 kJ/mol against a typical chemical 40–200) is exactly why pasteurisation is so temperature-sensitive: one extra degree buys a large factor in k (section 3.8, point 2).

Dimerisation in an MFR (slide 62). “Pure gaseous reactant A ($C_{A0} = 100$ milli mol/liter) is fed at a steady rate into a mixed flow reactor ($V = 0.1$ liter) where it dimerizes ($2A \rightarrow R$). For different gas feed rates the following data are obtained. Find a rate equation for this reaction.”

Run number	1	2	3	4
v_0 , liter/hr	10.0	3.0	1.2	0.5
C_{Af} , millimol/liter	85.7	66.7	50	33.4

Method (worked fully as Q13 in the companion document): for $2A \rightarrow R$ with pure A , $\varepsilon = y_{A0}\delta = -\frac{1}{2}$; solve $C_{Af} = C_{A0} \frac{1-X}{1+\varepsilon X}$ for X , then the MFR balance $-r_A = \frac{F_{A0}X}{V} = \frac{v_0 C_{A0}X}{V}$ gives one $(C_A, -r_A)$ pair per run:

$$X = 0.2502, 0.4996, 0.6667, 0.7995; \quad -r_A = 2502.2, 1498.9, 800.0, 399.8 \text{ mol L}^{-1}\text{h}^{-1}$$

A log–log fit gives $n = 1.96 \approx 2$, and dividing each run by C_A^2 :

$$\boxed{-r_A \approx 0.34 C_A^2}, \quad k = 0.3407, 0.3369, 0.3200, 0.3583 \text{ L mmol}^{-1}\text{h}^{-1}$$

The scatter in k across the four runs (0.320 to 0.358, about $\pm 6\%$) is the practical test that the assumed order is right: a wrong order makes k drift systematically instead of scattering (section 5 builds this test deliberately). The same numerical reappears in Ch5._krd.pdf (slide 77) and is solved once, as Q13.

Phosphine cracking in a PFR (slide 63). “The homogeneous gas decomposition of phosphine $4\text{PH}_3(g) \rightarrow \text{P}_4(g) + 6\text{H}_2(g)$ proceeds at 649 °C with the first-order rate $-r_{\text{PH}_3} = \left(\frac{10}{\text{hr}}\right) C_{\text{PH}_3}$. What size of plug flow reactor operating at 649 °C and 460 kPa can produce 80 % conversion of a feed consisting of 40 mol of pure phosphine per hour?” The slide's sketch labels the coil $4A \rightarrow R + 6S$ and asks for V .

Step 1 — inlet concentration at the reactor state (ideal gas):

$$C_{A0} = \frac{P}{RT} = \frac{460\,000}{8.314 \times 922.15} = 60.0 \text{ mol/m}^3.$$

Step 2 — expansion factor from the mole count as written, $4 \rightarrow (1 + 6) = 7$ moles per 4 reacted (P_4 is one mole of product molecules):

$$\delta = \frac{1}{4} + \frac{6}{4} - 0 - 1 = 0.75, \quad \varepsilon = y_{A0}\delta = 0.75 \quad (\text{pure } A).$$

Step 3 — PFR design equation, first order with $C_A = C_{A0}(1 - X)/(1 + \varepsilon X)$:

$$V = \frac{F_{A0}}{kC_{A0}} \int_0^X \frac{1 + \varepsilon X}{1 - X} dX = \frac{F_{A0}}{kC_{A0}} \left[(1 + \varepsilon) \ln \frac{1}{1 - X} - \varepsilon X \right].$$

Step 4 — numbers: $\frac{F_{A0}}{kC_{A0}} = \frac{40}{10 \times 60} = 0.0667 \text{ m}^3$; the bracket at $X = 0.8$: $1.75 \ln 5 - 0.75(0.8) = 2.817 - 0.6 = 2.217$:

$$\boxed{V = 0.0667 \times 2.217 = 0.148 \text{ m}^3 (\approx 148 \text{ L})}$$

Watch the mole count: P_4 is *one* mole of molecules (four atoms), so the products are $1+6 = 7$ moles and $\varepsilon = 0.75$. Counting atoms instead (“4 $\rightarrow$ 10”) inflates ε to 1.5 and the volume to 0.188 m^3 — the slip corrected here. The other classic slip is forgetting ε entirely ($V = 0.107 \text{ m}^3$, far too small) or evaluating C_{A0} at STP instead of at $649^\circ\text{C}/\text{kPa}$. Worked step by step as Q14 in the companion document.

4 Isothermal Reactor Design

Source: Ch4-Krd.pdf.

4.1 Conversion and the rate written in X (slides 5–6)

$$X = \frac{F_{A0} - F_A}{F_{A0}} \quad (\text{flow}), \quad X = \frac{N_{A0} - N_A}{N_{A0}} \quad (\text{batch}) \quad \implies \quad F_A = F_{A0}(1 - X) \text{ or } N_A = N_{A0}(1 - X)$$

$$-r_A = k C_{A0}^n \frac{(1 - X)^n}{(1 + \varepsilon X)^n} \quad (\text{power law combined with } C_A = C_{A0}(1 - X)/(1 + \varepsilon X), \text{ constant } T, P)$$

Where the X -form of the rate comes from. The rate is always evaluated at the *local* conditions. For an ideal-gas single reaction the flow table (§3.13–3.15) gives $F_A = F_{A0}(1 - X)$ and $v = v_0(1 + \varepsilon X)$ at constant T, P , so

$$C_A = \frac{F_A}{v} = \frac{F_{A0}(1 - X)}{v_0(1 + \varepsilon X)} = C_{A0} \frac{1 - X}{1 + \varepsilon X}.$$

Substituting into the power law $-r_A = k C_A^n$ gives the boxed form. If T or P vary, append the master ratios of §3.15:

$$C_A = C_{A0} \frac{1 - X}{1 + \varepsilon X} \cdot \frac{T_0}{T} \cdot \frac{P}{P_0}.$$

Express the rate in terms of X first; only then substitute $C_A(X)$. Slides 9–11 draw this as a loop (rate law $\rightarrow C_i$ in $X \rightarrow$ design equation $\rightarrow$ solve) and slide 12 packages it as the **five steps, in this order, every time**:

1. **Mole balance** — pick the reactor, write its design equation (section 1).
2. **Rate law** — from the data or the mechanism (section 3).
3. **Stoichiometry** — express every C_i in terms of X (the tables of §3.4).
4. **Combine** — substitute 2 and 3 into 1 to get one equation in X .
5. **Evaluate** — integrate, or solve, and size the reactor.

Slide 6 is a hand-drawn **Levenspiel plot**: $1/(-r_A)$ against X with a rectangle sketched to some target X . Its *area* (times F_{A0}) is the PFR volume $V = F_{A0} \int dX/(-r_A)$; its *height* at the exit is the CSTR integrand $V = F_{A0}X/(-r_A)|_{\text{exit}}$. Rectangle versus area is the entire PFR-versus-CSTR comparison in one picture (chapter 2). A final deck caution (slide 7): “elementary” is what licenses writing the rate law from the stoichiometry (mass-action); if a reaction is not elementary, the exponents come from data only — and the Arrhenius plot $\ln k$ versus $1/T$ is a straight line of slope $-E/R$ (section 3.8).

4.2 Closed-form conversions and the Damköhler number (slides 13–16, 25–28)

Batch, 1st order	$t = \frac{1}{k} \ln \frac{1}{1 - X} \quad (X = 1 - e^{-kt})$
Batch, 2nd order	$t = \frac{1}{kC_{A0}} \cdot \frac{X}{1 - X} \quad (X = \frac{kC_{A0}t}{1 + kC_{A0}t})$
CSTR, 1st order	$X = \frac{k\tau}{1 + k\tau}$
CSTR, 2nd order	$X = \frac{(1 + 2Da) - \sqrt{1 + 4Da}}{2Da}, \quad Da = k\tau C_{A0}$

Derivations (two lines each). Batch, constant volume: $t = C_{A0} \int_0^X dX/(-r_A)$ with $-r_A = kC_{A0}^n(1 - X)^n$ gives $\int_0^X (1 - X)^{-n} dX$: the logarithm for $n = 1$ and $\frac{1}{n-1} [(1 - X)^{1-n} - 1]$ otherwise — the two boxed rows (slide 15’s table also lists the n th-order result and the times for *half* life). CSTR: the balance $C_{A0}X = \tau k C_{A0}^n(1 - X)^n$ is *algebraic*: for $n = 1$, $X(1 + k\tau) = k\tau$; for $n = 2$, $Da X^2 - (1 + 2Da)X + Da = 0$, whose physical root (the one with $X \rightarrow 0$ as $Da \rightarrow 0$) is the minus branch of the quadratic formula (slide 28, “in terms of Da ” — the deck’s own typo).

$Da_1 = k\tau$	Damköhler number, first order
$Da \leq 0.1 \implies X < 0.1$	reaction-limited: little conversion
$Da \geq 10 \implies X > 0.9$	near-complete conversion

Da is the ratio of the reaction rate to the convective transport rate. It is the quickest way to see, before any integration,

whether a proposed reactor is remotely big enough.

4.3 Reactors in series and in parallel (slides 25–28)

$$N \text{ equal CSTRs in series, 1st order: } X = 1 - \frac{1}{(1 + k\tau_i)^N}$$

Parallel CSTRs: behave as one large CSTR of the same total volume

Series derivation (slide 25). One CSTR at steady state: $C_{\text{in}} - C_{\text{out}} = \tau k C_{\text{out}} \Rightarrow C_{\text{out}} = C_{\text{in}}/(1 + \tau k)$. Each tank multiplies the same factor, so after N tanks $C_N = C_0/(1 + \tau k)^N$ and $X = 1 - C_N/C_0$. Slide 26 plots X versus N at $Da = 0.1, 0.5, 1$: the gains are strongly diminishing, and the step from one tank to two is the biggest single improvement at fixed total volume. **Parallel:** m identical CSTRs sharing the feed have $\tau_i = V_i/v_0 = V_{\text{tot}}/(mv_0)$ each, and each solves $C_{\text{out}} = C_{\text{in}}/(1 + k\tau_i)$ with the *same* inlet — identical to one CSTR of volume V_{tot} (slide 27). Splitting changes the hydraulics, not the conversion.

Compare with §2.6: there, splitting the duty into two CSTRs saved 37% of the volume. This is the same result in closed form, and it is why industrial trains are staged rather than single-vessel.

4.4 Variable volume, gas phase (slides 38–46)

$$V = \int_0^X \frac{F_{A0} dX}{-r_A} \quad (\text{PFR}), \quad W = \int_0^X \frac{F_{A0} dX}{-r'_A} \quad (\text{PBR, per catalyst mass})$$

$$\text{Liquid } (\varepsilon = 0): \quad V = \frac{v_0}{k C_{A0}^{n-1}} \int_0^X \frac{dX}{(1 - X)^n} \quad \text{1st: } \frac{v_0}{k} \ln \frac{1}{1-X}; \quad \text{2nd: } \frac{v_0}{k C_{A0}} \frac{X}{1-X}$$

$$V_{\text{PFR}} = \frac{v_0}{k C_{A0}} \left[2\varepsilon(1 + \varepsilon) \ln(1 - X) + \varepsilon^2 X + \frac{(1 + \varepsilon)^2 X}{1 - X} \right] \quad \text{2nd order, gas (slide 42)}$$

Derivation of the gas 2nd-order result. With $C_A = C_{A0}(1 - X)/(1 + \varepsilon X)$ and $F_{A0} = v_0 C_{A0}$,

$$V = \frac{v_0}{k C_{A0}} \int_0^X \left(\frac{1 + \varepsilon X}{1 - X} \right)^2 dX,$$

and the decomposition $\left(\frac{1 + \varepsilon X}{1 - X} \right)^2 = \frac{(1 + \varepsilon)^2}{(1 - X)^2} - \frac{2\varepsilon(1 + \varepsilon)}{1 - X} + \varepsilon^2$ integrates term by term to the red box ($\int -\frac{2\varepsilon(1 + \varepsilon)}{1 - X} dX = 2\varepsilon(1 + \varepsilon) \ln(1 - X)$). Check the limit $\varepsilon \rightarrow 0$: the bracket collapses to $X/(1 - X)$, the liquid form. Slide 43 graphs the ε -dependence for $\varepsilon = -0.5, 0, 1, 2$: expansion ($\varepsilon > 0$) always inflates the required volume; contraction helps.

The first-order gas integral used in §4.10 below is the same procedure one order down: $\int_0^X \frac{1 + \varepsilon X}{1 - X} dX = (1 + \varepsilon) \ln \frac{1}{1 - X} - \varepsilon X$, which for $\varepsilon = 1$ is $2 \ln \frac{1}{1 - X} - X$ (the form in the ethane-cracking solution, slide 44). Slides 45–46 restate the master relations $C_A = C_{A0} \frac{1 - X}{1 + \varepsilon X} \frac{T_0}{T} \frac{P_0}{P}$ and $v = v_0(1 + \varepsilon X) \frac{T_0}{T} \frac{P_0}{P}$ — nothing new, but they are what makes the pressure-drop problem of §4.5–4.7 a *coupled* system.

4.5 Pressure drop in packed beds: Ergun, β_0 , and α (slides 47–51)

$$\frac{dX}{dW} = \frac{k C_{A0}^2}{F_{A0}} \left(\frac{1 - X}{1 + \varepsilon X} \right)^2 \left(\frac{P}{P_0} \right)^2 \quad \text{PBR, 2nd order with drop (slide 47)}$$

$$\frac{dP}{dz} = -\frac{G(1 - \phi)}{\rho_g g_c D_p \phi^3} \left[\frac{150(1 - \phi)\mu}{D_p} + 1.75 G \right] \quad \text{Ergun (slide 48)}$$

$$\beta_0 = \frac{G(1 - \phi)}{\rho_0 D_p \phi^3} \left[\frac{150(1 - \phi)\mu}{D_p} + 1.75 G \right] \quad (\text{slide 49})$$

$$W = z A_c \rho_b = z A_c (1 - \phi) \rho_c \quad \text{catalyst mass (slide 50)}$$

$$\frac{dP}{dW} = -\frac{\beta_0}{A_c(1 - \phi)\rho_c} \cdot \frac{P_0}{P} \cdot \frac{T}{T_0} \cdot \frac{F_T}{F_{T0}}$$

$$\alpha \equiv \frac{2\beta_0}{A_c(1 - \phi)\rho_c} \cdot \frac{1}{P_0} \quad \text{the lumped drop parameter}$$

From dP/dW to the reduced-pressure law. With $p = P/P_0$ and the slide's α ,

$$\frac{dp}{dW} = \frac{1}{P_0} \frac{dP}{dW} = -\frac{\alpha}{2p} \cdot \frac{T}{T_0} \cdot \frac{F_T}{F_{T0}} \xrightarrow{\text{single reaction: } F_T/F_{T0}=1+\varepsilon X} \frac{dp}{dW} = -\frac{\alpha}{2p} (1 + \varepsilon X) \quad (\text{constant } T).$$

For $\delta = 0$ (hence $\varepsilon = 0$) this *decouples* and integrates in closed form: $2p dp = -\alpha dW \Rightarrow p = (1 - \alpha W)^{1/2}$ — the formula of §4.6.

- ϕ is the bed porosity (void fraction), $(1 - \phi)$ the solid fraction, $G = \dot{m}/A_c$ the superficial mass flux, D_p the particle diameter, ρ_c the *solid* catalyst density and $\rho_b = (1 - \phi)\rho_c$ the *bulk* density. β_0 is the **Ergun equation** in the form the deck uses (viscous + inertial terms). The slide's full dP/dz form multiplies β_0 by the property corrections $\frac{P_0}{P} \frac{T}{T_0} \frac{F_T}{F_{T0}}$, and the English-unit form carries $g_c = 32.174 \text{ lb}_m \text{ ft}/(\text{lb}_f \text{ s}^2)$ (force–mass bridge).
- The grouping $A_c(1 - \phi)\rho_c$ in α is purely the conversion from bed *length* to catalyst *mass*: zA_c is the bed volume, $(1 - \phi)$ the solid volume fraction, ρ_c the solid density. Everything packing- and flow-specific sits in β_0 ; everything geometric or material sits outside.
- Because C_A carries a factor p (and for order n , p^n), pressure drop *slows the reaction*, so the conversion achieved in a given catalyst mass is lower than the no-drop calculation predicts. The two equations (X and p) must be solved **coupled**, not sequentially — except at $\delta = 0$, where the p -equation runs ahead on its own (§4.6).

4.6 PBR with $\delta = 0$: closed forms for X and p (slides 52–58)

$$p = (1 - \alpha W)^{1/2}, \quad \frac{X}{1 - X} = G^* \left(W - \frac{\alpha W^2}{2} \right), \quad G^* \equiv \frac{k C_{A0}^2}{F_{A0}} \quad (\text{2nd order})$$

no drop at all : $\frac{X}{1 - X} = G^* W.$

Derivation (the slide 53–57 sequence). At $\varepsilon = 0$ the rate is $-r_A = kC_{A0}^2(1 - X)^2 p^2$, so $\frac{dX}{dW} = G^*(1 - X)^2 p^2$ while $\frac{dp}{dW} = -\frac{\alpha}{2p}$ contains no X at all. Integrate the p -equation first ($p^2 = 1 - \alpha W$), substitute, separate:

$$\frac{dX}{(1 - X)^2} = G^*(1 - \alpha W) dW \quad \Rightarrow \quad \frac{X}{1 - X} = G^* \left(W - \frac{\alpha W^2}{2} \right).$$

The maximum useful catalyst mass is $W_{\max} = 1/\alpha$ (where $p \rightarrow 0$ and the rate dies) — the drag penalty grows quadratically in W while the benefit grows linearly.

Q18, the deck's board problem (slides 55–58): PBR for $A + B \rightarrow 2C$, isothermal, $W = 100 \text{ kg}$, $\alpha = 0.0099 \text{ kg}^{-1}$. The slides quote $k_A = 1.5 \text{ dm}^6 \text{ mol}^{-1} \text{ kg}^{-1} \text{ min}^{-1}$ and the results $X = 0.6$ (with drop) and $X = 0.75$ ($\alpha = 0$). The single lumped group that reproduces both is $G^* = kC_{A0}^2/F_{A0} = 0.03 \text{ kg}^{-1}$ (the value the deck's own numbers back-solve to; the individual k_A , C_{A0} , F_{A0} are not separately printed):

$$G^* \left(W - \frac{\alpha W^2}{2} \right) = 0.03 (100 - 49.5) = 1.515 \Rightarrow X = \frac{1.515}{2.515} = 0.602 \text{ (deck: 0.6)}, \quad G^* W = 3 \Rightarrow X = \frac{3}{4} = 0.75 \text{ exactly.}$$

Ignoring pressure drop overstates conversion by 0.15, a quarter of the achieved value. This is the number to quote when justifying a coupled solution.

4.7 The two-parameter (α, β) system and pipe friction (slides 59–63, 88–91)

$$\frac{dx}{dz} = \beta \frac{(1 - x)^n P^n}{(1 + \varepsilon x)^n}, \quad \frac{dP}{dz} = -\frac{\alpha}{P}, \quad \text{i.e.} \quad P^2 = P_{\text{in}}^2 - 2\alpha z$$

$$\beta = \frac{k A F_{A,\text{in}}^{n-1}}{P_{\text{in}}^n Q_{\text{in}}^n}, \quad \alpha_{\text{pipe}} = \frac{2f_F \rho_{\text{in}} u_{\text{in}}^2 P_{\text{in}}}{D} = \frac{32 f_F \rho_{\text{in}} Q_{\text{in}}^2 P_{\text{in}}}{\pi^2 D^5} \quad (\text{isothermal Fanning form, slides 89–91})$$

Derivation of β (slide 91's box). Isothermal ideal gas in terms of the volumetric inlet Q : $C_A = \frac{F_A}{Q} = \frac{F_{A,\text{in}}}{Q_{\text{in}}} \cdot \frac{P}{P_{\text{in}}} \cdot \frac{1-x}{1+\varepsilon x}$. With $-r_A = kC_A^n$ and $F_{A,\text{in}} dx/dz = A(-r_A)$,

$$\frac{dx}{dz} = \frac{kA F_{A,\text{in}}^{n-1}}{Q_{\text{in}}^n} \cdot \frac{(1-x)^n P^n}{P_{\text{in}}^n (1+\varepsilon x)^n},$$

which is the slide's β form (for $n = 1$: $\beta = kA/(P_{\text{in}}Q_{\text{in}})$).

Derivation of α_{pipe} . Fanning friction: $\frac{dP}{dz} = -\frac{2f_F \rho u^2}{D}$. At constant mass flux $\rho u^2 = \rho_{\text{in}} u_{\text{in}}^2 P/P_{\text{in}}$, so $P dP = -[2f_F \rho_{\text{in}} u_{\text{in}}^2 P_{\text{in}}/D] dz$ — exactly the P^2 -law above with α as boxed. (The deck's slide 90 collects the same objects in the Winterton pipe form and the Ergun β_0 form of §4.5; use whichever matches the data given.)

- **Series** beds share the same mass flow: their α 's add in the P^2 -law and the exit condition of one bed is the inlet of the next. **Parallel** channels split Q , so each sees smaller G and much less drop — but the deck's slides 74–75 conclude that shuffling beds between series and parallel under ΔP does not buy conversion: only total catalyst and total drop matter. Pumps and blowers (slides 61–63) are then sized from the integrated ΔP .
- **When does ΔP matter?** The deck's rule of thumb (slides 68–73): once the minor losses (fittings, valves, entrance/exit) are already a noticeable fraction (the slide uses $\sim 35\%$) of the pressure budget. Physically: an n th-order gas rate carries $(P/P_0)^n$ — halve the pressure and an $n = 2$ rate drops by four. If $\alpha W \ll 1$ (exit $p \approx 1$) pretend $\Delta P = 0$; otherwise integrate — analytically at $\delta = 0$ (§4.6), numerically otherwise (Q19, §4.13).

4.8 Worked from the deck: ethylene oxide in batch (Q15, slides 13–20)

The experiment (slide 18). Ethylene oxide hydrolysis ($\text{EO} + \text{water} \rightarrow \text{ethylene glycol}$, acid catalysed): **500 mL of 2 M EO mixed with 500 mL of 0.9 wt% aqueous H_2SO_4 at 55 °C** — the equal-volume dilution is already inside the table's C_{A0} — and the *product* (EG) concentration sampled over time. Slide 17 converts the data to (t, X) at $t = (0.5, 1, 1.5, 2, 3, 4, 6, 10)$ min; the deck's $\ln(1-X)$ column reads

$$(-0.157, -0.315, -0.472, -0.629, -0.942, -1.255, -1.884, -3.147).$$

First-order test: $\ln(1-X) = -kt$, so the least-squares slope of ℓ against t (its intercept fits at ≈ 0) is

$$k_{\text{fit}} = 0.315 \text{ min}^{-1} \quad (\ell_i = \ln(1-X)_i, t_i \text{ in min}),$$

and the batch time to 90 % is

$$t_{90} = \frac{\ln 10}{k} = 7.32 \text{ min}.$$

At the much slower $k = 10^{-4} \text{ s}^{-1}$ the same formula gives

$$t_{90\%} = \boxed{6.40 \text{ h}}$$

Slide 13–16 give the general batch algorithm these data are reduced with: Steps 1–5 of slide 12 plus the constant-volume integrals of §4.2 — $t_R = \frac{1}{kC_{A0}} \frac{X}{1-X}$ for 2nd order (so $t_R = 1/(kC_{A0})$, the time the *linear* extrapolation of the initial rate would need; see Q13 of the problem sheet) and $t = \frac{1}{k} \ln \frac{1}{1-X}$ for 1st order. Rate constant and units decide whether a batch is minutes or a working day: the factor 52 in t between $k = 0.315 \text{ min}^{-1}$ and 10^{-4} s^{-1} .

Two small discrepancies in the source, reported rather than smoothed over. (1) Slide 18's conversions at $t = 10$ min, $X = 0.995$, versus slide 19's $X = 0.957$ (the regression column is built from the latter). (2) The hand annotation on the fit, $k = 0.311 \text{ min}^{-1}$, versus the least-squares slope $k_{\text{fit}} = 0.315 \text{ min}^{-1}$. Any exam answer in $[0.31, 0.32] \text{ min}^{-1}$ is consistent with the slides.

4.9 Worked from the deck: ethylene glycol in CSTR networks (Q16, slides 29–38)

Problem data (slides 29–32). Aqueous EO hydrolysis, pseudo-first-order in EO with $k = 0.311 \text{ min}^{-1}$ at 55 °C (the value used throughout the worked solution). The plant must make **200 million lb of ethylene glycol per year**. The EO feed is a 1 lbmol/ft³ solution mixed with an *equal volume* of 0.9 wt% aqueous H_2SO_4 ; $MW_{\text{EG}} = 62 \text{ lb/lbmol}$. **800-gal CSTRs** are available; the year is 525 600 min and the plant runs at 80 % of theoretical capacity.

The five steps, with numbers (slides 30–38).

- Step 1** Product spec $\rightarrow$ molar production: $F_C = 200 \times 10^6 \frac{\text{lb}}{\text{yr}} \times \frac{1 \text{ lbmol}}{62 \text{ lb}} \times \frac{1 \text{ yr}}{525600 \text{ min}} = 6.137 \text{ lbmol/min}$ (the deck’s “80 % of theoretical capacity” is the design conversion $X = 0.8$ of Step 2 — theoretical capacity is $X = 1$).
- Step 2** Stoichiometry ($\text{EO} \rightarrow \text{EG}$ is 1:1) with $X = 0.8$: $F_{A0} = F_C/0.8 = 7.672 \text{ lbmol/min}$ entering the reactor.
- Step 3** After 1:1 mixing, $C_{A0} = 0.5 \text{ lbmol/ft}^3$ and $v_0 = F_{A0}/C_{A0} = 15.34 \text{ ft}^3/\text{min}$.
- Step 4** One CSTR at $X = 0.8$: $V = \frac{F_{A0}X}{kC_{A0}(1-X)} = \frac{7.672 \times 0.8}{0.311 \times 0.5 \times 0.2} = \boxed{197.3 \text{ ft}^3}$.
- Step 5** Use the 800-gal tanks (800 gal = 106.9 ft³ each). Two in *parallel*: $\tau = 2 \times 106.9/15.34 = 13.94 \text{ min}$, $Da = 4.34$, $X = 1 - \frac{1}{1+Da} = 0.813$ (deck: 0.81). Two in *series*: each $\tau = 6.97 \text{ min}$, $Da = 2.168$, $X = 1 - \frac{1}{(1+Da)^2} = 0.900$ (deck: 0.90).

Arrangement	τ (min)	Da	X
two CSTRs in parallel	13.94	4.34	0.813
two CSTRs in series	6.97	2.168	0.900

Same total volume, and the series arrangement reaches 0.900 against 0.813 — the staging argument of §4.3 with the deck’s own numbers. Two bookkeeping cautions from the slides: the “80 %” is one number used twice — the same design conversion $X = 0.8$ in $F_{A0} = F_C/X$ (Step 2) and in the CSTR balance (Step 4) — not an extra capacity factor to apply again; and the pseudo-first-order k lumps the acid catalysis (the same chemistry as Q15 at $k \approx 0.31 \text{ min}^{-1}$).

4.10 Worked from the deck: ethane cracking in a PFR (Q17, slides 44, 85–88)

Problem (slide 44). Thermal cracking $\text{C}_2\text{H}_6 \rightarrow \text{C}_2\text{H}_4 + \text{H}_2$, first order, $k = 0.072 \text{ s}^{-1}$ at 1000 K, $E = 82000 \text{ cal/mol}$ (82 kcal/mol — the deck’s MATLAB uses $R = 1.987 \text{ cal mol}^{-1}\text{K}^{-1}$, slide 88). Feed 300 M lb/yr of pure ethane at 6 atm; reactor at 1100 K; $X = 0.8$. Tubes of $L = 40 \text{ ft}$, ID = 1 in. **How many tubes?**

Step 1 — kinetics: $k_{1100} = 0.072 \exp\left(\frac{82000}{1.987}\left(\frac{1}{1000} - \frac{1}{1100}\right)\right) = 3.067 \text{ s}^{-1}$ — the Arrhenius step moves k by a factor of 43 for 100 K.

Step 2 — feed: $F_{A0} = \frac{300 \times 10^6 \text{ lb/yr}}{28 \text{ lb/lbmol}} \times \frac{1}{3.1536 \times 10^7 \text{ s/yr}} \times 453.6 \text{ mol/lbmol} = 154.1 \text{ mol/s}$; $C_{A0} = \frac{P}{RT} = \frac{6 \times 101325}{8.314 \times 1100} = 66.476 \text{ mol/m}^3$; $v_0 = F_{A0}/C_{A0} = 2.318 \text{ m}^3/\text{s}$.

Step 3 — expansion factor: $\text{A} \rightarrow \text{B} + \text{C}$ is $\delta = +1$ with $y_{A0} = 1$ (the slide 88 code writes it as `epsilon = yao*(1+1-1)`), so $\varepsilon = 1$ and $V = \frac{v_0}{k} \left[2 \ln \frac{1}{1-X} - X \right]$ (§4.4).

Step 4 — size: at $X = 0.8$: $V = \frac{2.318}{3.067} [2 \ln 5 - 0.8] = 0.7558 \times 2.4183 = \boxed{1.83 \text{ m}^3 = 64.6 \text{ ft}^3}$.

Step 5 — tube count: one tube is $\frac{\pi}{4} \left(\frac{1}{12} \text{ ft}\right)^2 \times 40 \text{ ft} = 0.218 \text{ ft}^3$, so $N = 64.6/0.218 = \mathbf{296}$ tubes (about 300 with design margin).

A units trap worth knowing (the “3150 tubes” figure). Some worked solutions give 3150 tubes of 0.0205 ft³ each. That 0.0205 is $\frac{\pi}{4}(0.0254)^2 \times 40$ — a metric area (m²) multiplied by a feet length (ft) — divided into 64.6 ft³. With consistent units the tube is 0.218 ft³ and the count is **296**. Sanity check: $N \times V_{\text{tube}}$ must return the total V in the same units.

Slide 88 gives the companion MATLAB for the rate profile along the same reactor (`Vspan = [0 2310]; y0 = [0]; ode45(@ODEfun,Vspan,y0);` with `Ca(i) = Cao*(1-y(i,1))/(1+epsilon*y(i,1)); ra(i) = -k*Ca(i);`, plotted as `plot(v,ra), xlabel('Volume'), ylabel('-ra')`). Note the code’s `Cao = 0.066` is C_{A0} in kmol/m³ — the same 66.476 mol/m³ rounded.

4.11 Worked from the deck: packed bed, $\text{A} + \text{B} \rightarrow 2\text{C}$ (Q18, slides 55–58)

The board solution is §4.6’s closed form with the deck’s numbers: $W = 100 \text{ kg}$, $\alpha = 0.0099 \text{ kg}^{-1}$, $G^* = 0.03 \text{ kg}^{-1}$ (back-solved; the slide quotes $k_A = 1.5 \text{ dm}^6 \text{ mol}^{-1} \text{ kg}^{-1} \text{ min}^{-1}$ without the individual C_{A0} , F_{A0}):

$$\boxed{X = 0.602 \text{ with pressure drop}} \quad \text{vs.} \quad \boxed{X = 0.75 \text{ without}}$$

Note also the slide’s $\delta = 0$ pressure profile $p = (1 - \alpha W)^{1/2}$ and the rate carried in the form $-r_A = kC_{A0}^2(1-X)^2(1-\alpha W)$ (slide 57) — the drop enters the *rate*, not just the size.

Ignoring pressure drop overstates conversion by 0.15, a quarter of the achieved value. This is the number to quote when justifying a coupled solution.

4.12 Worked from the deck: batch series $\text{A} \rightarrow \text{B} \rightarrow \text{C}$ (Q20, slides 77–79)

Problem (slide 78). The isothermal series reaction $A \rightarrow B \rightarrow C$ runs in a batch reactor with $k_1 = 1 \text{ h}^{-1}$, $k_2 = 2 \text{ h}^{-1}$, and at $t = 0$: $C_A = 5$, $C_B = C_C = 0$ (the slide prints mol/m^3). Solve for the three concentrations and plot them versus time.

The ODE system the deck codes (MATLAB `ode45`, slides 78–79):

$$\frac{dC_A}{dt} = -k_1 C_A, \quad \frac{dC_B}{dt} = k_1 C_A - k_2 C_B, \quad \frac{dC_C}{dt} = k_2 C_B.$$

Exact solution: $C_A = C_{A0} e^{-k_1 t}$, $C_B = C_{A0} \frac{k_1}{k_2 - k_1} (e^{-k_1 t} - e^{-k_2 t})$, $C_C = C_{A0} - C_A - C_B$. **The intermediate maximum:** set $dC_B/dt = 0$:

$$\frac{k_1}{k_2 - k_1} (-k_1 e^{-k_1 t} + k_2 e^{-k_2 t}) = 0 \implies e^{(k_2 - k_1)t} = \frac{k_2}{k_1} \implies t_{\max} = \frac{\ln(k_2/k_1)}{k_2 - k_1} = \boxed{0.693 \text{ h}},$$

and back-substituting,

$$C_{B,\max} = C_{A0} \left(\frac{k_1}{k_2} \right)^{k_2/(k_2 - k_1)} = 5 \times (0.5)^2 = \boxed{1.250}$$

(B reaches only a quarter of the initial A — with $k_2 > k_1$ the intermediate is destroyed about as fast as the first step can make it; for a *selective* series route one wants $k_2 \ll k_1$).

The deck's plotting block (slide 79): `plot(t,c(:,1),'+',t,c(:,2),'*',t,c(:,3)), legend('ca','cb','cc'), xlabel('time(hr)'), ylabel('concs, mol/m3'), title('Batch reactor'))`. The printed $C_{A0} = 5 \text{ mol/m}^3$ is dimensionally odd for a liquid batch but is the scale the code and plot use; every *shape* and both boxed results scale linearly or not at all with it ($C_{B,\max}$ is linear in C_{A0} , $t_{\max}$ independent). The `ode45('batch',...)` call on slide 79 is old MATLAB string syntax; modern versions want the handle `@batch`.

4.13 Worked from the deck: CSTR and PBR in either order (Q19, slides 74–77)

Problem (slide 76). The irreversible gas-phase reaction $A \rightarrow B$, **first order in the partial pressure of A**, runs isothermally in a **fluidized** catalytic reactor holding 50 kg of catalyst. At present $X = 0.50$ is realized with pure A entering at 20 atm — *there is virtually no pressure drop in the CSTR*. It is proposed to put a PBR of the same catalyst weight in series with the CSTR. For the PBR, $\alpha = 0.018 \text{ kg}^{-1}$ (particle size 0.2 mm, bed porosity 40 %, viscosity as air at 200 °C). (a) PBR upstream or downstream of the CSTR for the highest exit conversion? (b) The exit conversion of the last reactor? (c) The pressure at the exit of the packed bed? (d) How do the answers change if the catalyst diameter is halved and the PBR diameter increased by 50 %, assuming turbulent flow?

Model (read the slide literally). $A \rightarrow B$ has $\delta = 0$, so $\varepsilon = 0$ and $p_A = P(1 - X)$: the rate $-r'_A = k' p_A = k' P_0(1 - X)p$ gives the coupled pair

$$\frac{dX}{dW} = a(1 - X)p, \quad \frac{dp}{dW} = -\frac{\alpha}{2p}, \quad a \equiv \frac{k' P_0}{F_{A0}} [\text{kg}^{-1}].$$

Calibration from the existing plant: the fluidized reactor *is* the CSTR (no drop, $p = 1$), so $W = F_{A0} X / (k' P_0(1 - X))$ at $X = 0.50$, $W = 50 \text{ kg}$ fixes

$$a = \frac{X}{W(1 - X)} = \frac{0.5}{50 \times 0.5} = 0.0200 \text{ kg}^{-1} \quad (\text{exact}).$$

(c) At $\delta = 0$ the exit pressure is model-independent: $p_{\text{exit}} = (1 - \alpha W)^{1/2} = (1 - 0.9)^{1/2} = 0.3162$, i.e. $P = 20 \times 0.3162 = \boxed{6.32 \text{ atm}}$.

(a, b) RK-4 integration of the pair ($h = 0.05 \text{ kg}$):

- **CSTR then PBR** (PBR downstream): the CSTR delivers $X = 0.5$ at $p = 1$; the PBR continues to $X = \mathbf{0.756}$ (its own exit $p = 0.3162$).
- **PBR then CSTR:** the PBR alone reaches $X = 0.512$ at $p = 0.3162$; the trailing CSTR must then work at 32 % of full pressure and finishes at $X = \mathbf{0.629}$.

Answer (a): the PBR goes downstream (CSTR first). (b): $X_{\text{exit}} = 0.756$.

(d) From the deck's own $\alpha = 2\beta_0/[A_c(1 - \phi)\rho_c P_0]$ with turbulent Ergun $\beta_0 \propto G^2/D_p$ and $G = \dot{m}/A_c$:

$$\alpha \propto \frac{1}{A_c^3 D_p}.$$

$D_p \rightarrow D_p/2$ gives $\times 2$; tube diameter $\times 1.5 \Rightarrow A_c \times 2.25$ gives $\times 2.25^{-3}$:

$$\alpha' = 0.018 \times \frac{2}{2.25^3} = 0.00316 \text{ kg}^{-1}.$$

Re-integrating: CSTR then PBR $X = 0.808$ (from 0.756); PBR then CSTR $X = 0.800$ (from 0.629); the two arrangements nearly converge and the packed-bed exit pressure rises to $p = 0.918$ (18.4 atm).

The α -scaling of part (d) is easy to half-miss. The friction group β_0 itself sees only G^2/D_p ($\times 2$ and $\times 2.25^{-2}$ would give $\alpha' = 0.00711 \text{ kg}^{-1}$), but converting the pressure gradient to a drop *per mass of catalyst* contributes the extra $1/A_c$ of the wider, shorter bed — the full 2.25^{-3} and $\alpha' = 0.00316 \text{ kg}^{-1}$. And note (c) is purely geometric: $p = \sqrt{1 - \alpha W}$ carries no kinetics at all.

Arrangement	$a \text{ (kg}^{-1}\text{)}$	X_{exit}
PBR then CSTR	0.0200	0.629
CSTR then PBR	0.0200	0.756
PBR then CSTR, $\alpha' = 0.00316$	0.0200	0.800
CSTR then PBR, $\alpha' = 0.00316$	0.0200	0.808

Why CSTR-first wins: the CSTR converts at the *inlet* pressure and its rate is first order in p ; the PBR then scavenges the remainder in plug flow with gradually falling p . In the reverse order the PBR burns its volume near $X \approx 0$ (fast) but exits at $p = 0.32$, and the CSTR is stuck at that low pressure for its whole volume. The ordering is worth 0.13 of conversion at $\alpha = 0.018 \text{ kg}^{-1}$ — and washes out once the drop is nearly removed (part d: difference 0.008). All numbers reproduced by `krd-answers-check.py`.

4.14 Worked from the deck: pressure-drop length check (Q21, slides 88–99)

Problem data (slides 89–91). An isothermal gas-phase isomerization (so $\varepsilon = 0$) in a tube of $D = 2 \text{ cm}$ with $Q_{\text{in}} = 0.005 \text{ m}^3/\text{s}$ at $P_{\text{in}} = 10 \text{ atm}$, $\mu = 10^{-5} \text{ kg}/(\text{m}\cdot\text{s})$, $T = 300 \text{ K}$, $M = 60 \text{ g/mol}$, $R = 0.08206 \text{ atm L (gmol K)}^{-1}$; first order in C_A with $k = 0.1 \text{ (m}^3/\text{s)}/\text{m}^3$. The deck's questions: **(a)** the pressure drop through the reactor; **(b, c)** the lengths for $X = 0.10$ and $X = 0.20$ at $\Delta P = 0$; **(d)** the length to reach $X = 0.20$ *accounting for the drop* — and the later slides ask in particular what is reachable in a 1.5 cm variant.

Slide 90–91's printed working:

$$\rho_{\text{in}} = \frac{P_{\text{in}} M}{RT} = 24.0558 \text{ kg/m}^3, \quad u_{\text{in}} = \frac{Q_{\text{in}}}{\pi D^2/4} = 15.9155 \text{ m/s}, \quad F_{A,\text{in}} = 2.0047 \text{ mol/s},$$

$$Re = \frac{\rho_{\text{in}} u_{\text{in}} D}{\mu} = 7.65 \times 10^5 \text{ (turbulent)}, \quad f = 0.003 \text{ (Fanning)},$$

$$\alpha = 1.847 \times 10^9 \text{ Pa}^2/\text{m} \text{ (deck's value; the §4.7 formula gives } 2f_F \rho_{\text{in}} u_{\text{in}}^2 P_{\text{in}}/D = 1.83 \times 10^9 \text{ Pa}^2/\text{m, within about 1\%)},$$

$$\beta = \frac{kA}{P_{\text{in}} Q_{\text{in}}} = 6.2832 \times 10^{-9} \text{ 1/(Pa m)} \quad (n = 1 \text{ case of slide 91's box}).$$

(b, c) At $\Delta P = 0$: $p = 1$ and $dx/dz = \beta(1-x)P_{\text{in}} \equiv k_v(1-x)$ with $k_v = \beta P_{\text{in}} = 6.283 \times 10^{-3} \text{ m}^{-1} = k/u_{\text{in}}$, so $x = 1 - e^{-k_v z}$:

$$L(10\%) = \frac{\ln(1/0.9)}{k_v} = \boxed{16.8 \text{ m}}, \quad L(20\%) = \frac{\ln(1/0.8)}{k_v} = \boxed{35.5 \text{ m}}.$$

(a, d) With the drop: $p = (1 - z/z_c)^{1/2}$ with $z_c = P_{\text{in}}^2/(2\alpha) = 270.7 \text{ m}$ and $dx/dz = k_v(1-x)p$ integrate in closed form:

$$x(z) = 1 - \exp\left(-\frac{2}{3}k_v z_c \left[1 - (1 - z/z_c)^{3/2}\right]\right).$$

It gives $x = 0.10$ at $z = 16.9 \text{ m}$ (the deck's “expanded” plot reads $\approx 17 \text{ m}$) and $x = 0.43$ at $z = 100 \text{ m}$ with $P = 7.9 \times 10^5 \text{ Pa}$ (the deck's plots print 0.43 and 7.8×10^5) — mild drop for the 2 cm tube. Slides 98–99 rerun the case at 1.5 cm and annotate in red: **“10% conversion is possible, but 20% is not”** — x saturates near 0.15 (a narrower tube raises the turbulent $\alpha \propto 1/D^5$ by $(2/1.5)^5 = 4.2\times$).

A discrepancy in the source, reported rather than smoothed over. The deck states that about 20 m achieves the target *with* pressure drop (the closed form says 16.9 m), and asks for $X = 0.20$ in a 1.5 cm bed; its own plots answer “10% conversion is possible, but 20% is not” with x saturating near 0.15. Two tensions inside those figures are kept visible rather than smoothed: (1) in the 2 cm run the closed form reaches $X = 0.20$ at $z = 36.8 \text{ m}$ (and 0.43 at 100 m), so the “20% is not possible” is a statement about the 1.5 cm case only; (2) a pure $\alpha \propto 1/D^5$ rescaling would let the 1.5 cm tube reach 0.20 near 45 m and saturate near 0.24 — against the deck's own run (saturation ≈ 0.15 , “20% is not possible”). The deck's run is the figure to quote; the gap is flagged here rather than resolved by adjusting a number.

Two readings notes. The printed $R = 0.08206 \text{ atm L (gmol K)}^{-1}$ with exactly 10 atm would give $\rho_{\text{in}} = 24.37 \text{ kg/m}^3$, but the printed 24.0558 matches $P_{\text{in}} = 10^6 \text{ Pa}$ exactly — the deck treats “ 10 atm ” as 10 bar throughout (same in α). Quote the printed $\alpha = 1.847 \times 10^9 \text{ Pa}^2/\text{m}$ and $\beta = 6.2832 \times 10^{-9} \text{ 1/(Pa m)}$ in an exam; the §4.7 formulas reproduce them to $\sim 1\%$ and exactly, respectively. All numbers above are reproduced by `krd-answers-check.py`.

5 Collection and Analysis of Rate Data

Source: Ch5_–_krd.pdf.

5.1 Differential method

- Take logs of the power law to linearise it:

$$\ln\left(-\frac{dC_A}{dt}\right) = \ln k + \alpha \ln C_A \quad \text{slope} = \alpha, \text{ intercept} = \ln k$$

The derivatives come from the concentration–time data by finite differences:

initial point	$\frac{-3C_0 + 4C_1 - C_2}{2\Delta t}$
interior point	$\frac{C_{i-1} - C_{i+1}}{2\Delta t}$
final point	$\frac{C_{N-2} - 4C_{N-1} + 3C_N}{2\Delta t}$

5.2 Integral method

- **Guess the order, integrate, and test for a straight line.** The order that linearises the data is the order.

order 0	C_A against t is linear
order 1	$\ln C_A$ against t is linear
order 2	$1/C_A$ against t is linear
$A + B, M = C_{B0}/C_{A0} \neq 1$	$\ln \frac{C_A}{C_B} = \ln M + C_{A0}(M - 1)kt$

5.3 Half-life method

$t_{1/2} = \frac{2^{n-1} - 1}{(n-1)k} C_{A0}^{1-n}$	order $n \neq 1$
$\ln t_{1/2} = (1-n) \ln C_{A0} + \text{const}$	slope $= 1-n \implies n = 1 - \text{slope}$
$n = 1 : t_{1/2} = \frac{\ln 2}{k}$	independent of C_{A0}

A half-life that does not change with initial concentration is the signature of first-order kinetics — no regression needed.

5.4 Differential reactor and initial rates

$$-r'_A = \frac{F_{A0} - F_A}{W} \approx \frac{F_P}{W} \quad \text{low conversion, so composition is nearly uniform}$$

5.5 Nonlinear regression and enzyme kinetics

$s^2 = \sum_i (C_{m,i} - C_{c,i})^2$	minimise; m = measured, c = calculated
$-r_S = \frac{k_3 C_S C_E}{C_S + C_M}$	Michaelis–Menten
$k_3 C_E t = (C_{S0} - C_S) + C_M \ln \frac{C_{S0}}{C_S}$	integrated form

5.6 Variable-volume batch, constant pressure

$$X = \frac{(V/V_0) - 1}{\varepsilon_A} \quad \text{read conversion straight off the volume change}$$

5.7 Worked from the deck

Problem	Result
Trityl + methanol (regression of $\ln t_{1/2}$)	$n = 2.14$; $k' = 0.1243$; $k = 0.249 \text{ dm}^6\text{mol}^{-2}\text{min}^{-1}$
Dimethyl ether, constant volume	$p_{i0} = 310.3$; $k = 4.4, 4.3, 4.4, 4.9 \times 10^{-4} \text{ s}^{-1}$, mean 4.50×10^{-4}
$A \rightleftharpoons R$, reversible	$k_1 = 0.05776$, $k_2 = 0.02888$
L5.19	$n = 2.00$; $k = 0.0040, 0.0040, 0.00402, 0.00401 \text{ L mmol}^{-1}\text{min}^{-1}$
L3.25	$k = 0.1604 \text{ min}^{-1}$
Bomb reactor, $2A \rightarrow B$	1st order $k = 0.0517$, $r^2 = 0.886$; 2nd order $k = 0.0741$, $r^2 = 0.964$
Six MFRs in series vs one PFR	$X_{\text{MFRs}} = 0.750$, $X_{\text{PFR}} = 0.667$
Two-reactor split	$X_A = 0.2308$, $X_B = 0.1154$, $C_B = 46.00$
Sucrose inversion, Michaelis–Menten	$k_3 C_{S0} = 0.3210 \text{ h}^{-1}$; $C_M = 0.5151 \text{ mmol/L}$; $k_3 = 32.1 \text{ h}^{-1}$

- **The bomb-reactor comparison is the cleanest illustration of the integral method in the whole deck.** Both orders were fitted to the same data; the second-order fit has $r^2 = 0.964$ against 0.886 for first order. The higher r^2 is what selects the order — not a preference, and not the value of k .

- **Reversible PFR** (slide 61), $A \rightleftharpoons R$:

$$X_e = 0.80, \quad \tau = 20.0 \text{ min}, \quad C_{Af} = 0.0494 \quad \Longrightarrow \quad \boxed{X = 0.506}$$

$X = 0.506$ against an equilibrium ceiling of $X_e = 0.80$: the reactor is approaching equilibrium, not running out of residence time. Adding volume beyond this point buys very little, which is the practical significance of computing X_e first.

6 Multiple Reactions

Source: Ch6_krd.pdf.

6.1 The four classes of multiple reaction

- **Parallel (competing):** $A \xrightarrow{k_1} B$ and $A \xrightarrow{k_2} C$ consume the same feedstock at once; the reactor's job is to steer consumption toward one branch.
- **Series (consecutive):** $A \xrightarrow{k_1} B \xrightarrow{k_2} C$ — the product of the first reaction is the feed of the second, so the intermediate B passes through a maximum.
- **Complex (parallel + series):** e.g. $A + B \xrightarrow{k_1} C + D$ together with $A + C \xrightarrow{k_2} E$; the deck also prints the **Denbigh network** $A \xrightarrow{1} R \xrightarrow{3} S$ with the side arrows $A \xrightarrow{2} T$ and $R \xrightarrow{4} U$ — the standard bookkeeping test for net rates.
- **Independent:** $A \rightarrow B + C$ and $D \rightarrow E + F$ run side by side without sharing species.

The deck's own chemistry: parallel — ethylene + oxygen burning to $2\text{CO}_2 + 2\text{H}_2\text{O}$ (k_1) versus epoxidation to ethylene oxide (k_2); series — ethylene oxide + NH_3 to mono-, then di-, then tri-ethanolamine; complex — ethanol to ethylene and to acetaldehyde, then ethylene + acetaldehyde to butadiene + water; independent — cracking of crude, $\text{C}_{15}\text{H}_{32} \rightarrow \text{C}_{12}\text{H}_{26} + \text{C}_3\text{H}_6$ and $\text{C}_8\text{H}_{18} \rightarrow \text{C}_6\text{H}_{14} + \text{C}_2\text{H}_4$.

With two reactants the deck's first lever is the *contacting pattern* (slide 5):

- C_A, C_B both high — charge A and B all at once (batch);
- C_A, C_B both low — add both slowly to a well-mixed vessel (semibatch / fed CSTR);
- C_B high, C_A low — start with B and feed A slowly.

The economic incentive (slide 8): with D desired and U undesired, maximise D and minimise U ; the deck's cost sketch shows the reaction cost falling and the separation cost rising as the system is pushed harder, so the *total* cost has a minimum in between — selectivity, not conversion alone, sets the optimum.

6.2 Selectivity and yield — the four definitions

$S_{D/U} = \frac{r_D}{r_U}$	instantaneous (rate) selectivity
$\tilde{S}_{D/U} = \frac{F_D}{F_U}, \frac{N_D}{N_U}$ (batch)	overall selectivity
$Y_D = \frac{r_D}{-r_A}$	instantaneous yield
$\tilde{Y}_D = \frac{F_D}{F_{A0} - F_A}, \frac{N_D}{N_{A0} - N_A}$ (batch)	overall yield

- In a CSTR the two selectivities coincide (slide 11): $F_D = r_D V$ and $F_U = r_U V$, so $\tilde{S}_{D/U} = r_D V / r_U V = S_{D/U}$.
- Definitions differ between authors — slide 14 says to check which one a question intends. The *overall* quantities set the profit of the reaction–separation system; the *instantaneous* ones are the design tool, because they say which reactor and contacting scheme to choose.

6.3 The rate selectivity parameter, one reactant

For $A \xrightarrow{k_D} D$ (desired) and $A \xrightarrow{k_U} U$ (undesired) with $-r_A = k_D C_A^{\alpha_1} + k_U C_A^{\alpha_2}$:

$S_{D/U} = \frac{r_D}{r_U} = \frac{k_D}{k_U} C_A^{\alpha_1 - \alpha_2}$	maximise this
$\frac{k_D}{k_U} = \frac{A_D}{A_U} e^{-(E_D - E_U)/RT}$	the temperature lever

- | | |
|-------------------------------------|---|
| Case 1 $\alpha_1 > \alpha_2$ | keep C_A high : batch or plug flow, no inerts and high pressure (gas), minimum diluent (liquid) |
| Case 2 $\alpha_2 > \alpha_1$ | keep C_A low : CSTR or recycle reactor (the product stream acts as diluent), inerts and low pressure (gas), maximum diluent (liquid) |
| Case 3 $E_D > E_U$ | run hot : k_D grows faster with T than k_U |
| Case 4 $E_U > E_D$ | run cool — but not so cool that the desired reaction stops proceeding |

The deck's rules of thumb (slide 4) are the same table in words: a *high* reactant concentration favours the reaction of *higher* order, a *low* concentration the reaction of *lower* order, and for reactions of the *same* order concentration has no effect on the distribution.

- **Worked selectivity argument** (slides 18–19). Three parallel paths from A :

$$r_D = 0.0002 e^{36\,000(\frac{1}{300} - \frac{1}{T})} C_A \quad r_U = 0.0018 e^{25\,000(\frac{1}{300} - \frac{1}{T})} C_A^{1.5} \quad r_Q = 0.00452 e^{5\,000(\frac{1}{300} - \frac{1}{T})} C_A^{0.5}$$

$S_{D/Q} = \frac{r_D}{r_Q} \approx \text{very large at high } T$	$(E_D = 36\,000 \gg E_Q = 5\,000)$
$S_{D/U} = \frac{r_D}{r_U} = \frac{0.11 e^{11\,000(\frac{1}{300} - \frac{1}{T})}}{C_A^{0.5}}$	$(\frac{0.0002}{0.0018} = 0.11, \quad 36\,000 - 25\,000 = 11\,000, \quad 1 - 1.5 = -\frac{1}{2})$

Two levers survive the algebra: **high temperature** (to bury Q) and **low concentration of A** (to bury U) — the latter by inerts, low pressure, or a CSTR / recycle reactor. One printed sentence runs against the algebra: slide 18's " S_{DU} is minimized at low concentration of A " states the reverse of the $C_A^{-0.5}$ dependence of the slide's own equation (5); slide 19's operating conclusions agree with the algebra, and the sentence is reported as printed.

6.4 Two reactants: signs, schemes and the four cases

For $A + B \xrightarrow{k_1} D$ and $A + B \xrightarrow{k_2} U$ with $r_D = k_1 C_A^{\alpha_1} C_B^{\beta_1}$, $r_U = k_2 C_A^{\alpha_2} C_B^{\beta_2}$:

$$S_{D/U} = \frac{k_1}{k_2} C_A^{\alpha_1 - \alpha_2} C_B^{\beta_1 - \beta_2}$$

The deck's menu of contacting schemes (slide 21): (a) CSTR; (b) tubular reactor; (c) batch; (d) semibatch, B fed slowly into A; (e) semibatch, A fed slowly into B; (f) tubular reactor with side streams of B; (g) tubular reactor with side streams of A; (h) series of small CSTRs; (i) tubular reactor with recycle.

Case I $\alpha_1 > \alpha_2, \beta_1 > \beta_2$	keep C_A, C_B both high — (b) tubular, (c) batch, high pressure, reduce inert
Case II $\alpha_1 > \alpha_2, \beta_1 < \beta_2$	C_A high, C_B low — (d) semibatch B into A, (f) side streams of B, (h) small CSTRs with A fed only to the first
Case III $\alpha_1 < \alpha_2, \beta_1 < \beta_2$	keep C_A, C_B both low — (a) CSTR, (i) large recycle ratio, feed diluted with inert, low pressure
Case IV $\alpha_1 < \alpha_2, \beta_1 > \beta_2$	C_A low, C_B high — (e) semibatch A into B, (g) side streams of A, (h) small CSTRs with fresh B to each

A slip in the source, reported rather than corrected. Slide 25 (Case IV) opens with the sentence “maintain the concentration of both A and B as high as possible,” which contradicts its own $S_{D/U} = k_1 C_B^b / (k_2 C_A^a)$ with $a, b > 0$; the schemes it then lists — (e), (g), (h) with fresh B — are exactly the low- C_A /high- C_B set quoted in the table above.

6.5 Series reactions — maximising the intermediate

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C \quad \Rightarrow \quad C_B = C_{A0} \frac{k_1}{k_2 - k_1} (e^{-k_1 t} - e^{-k_2 t})$$

$$t_{\text{opt}} = \frac{\ln(k_2/k_1)}{k_2 - k_1} \quad \text{maximises } C_B$$

- In **parallel** networks the product is maximised by adjusting the reaction *conditions* and choosing the proper reactor; in **series** networks the lever is *time* — the space time of a flow reactor or the real time of a batch. The $C_B(t)$ form above is the one behind the Ch4 series numerical reproduced in §4.12 (Q20).
- The deck's triangular diagram (slide 27): $k_1/k_2 \gg 1$ gives a large quantity of B; $k_1/k_2 \sim 1$ a moderate one; $k_1/k_2 \ll 1$ (first reaction slow, second fast) very little. In every case a *long* reaction time in batch or a *long* tubular reactor converts B onward to C — which is why the optimum is interior.
- Verified numerically in §4.12: $t_{\text{max}} = 0.693$ h gives $C_{B,\text{max}} = 1.250$.

6.6 Product distribution — the quantitative treatment

$$\varphi = \frac{dC_R}{-dC_A} \quad \text{instantaneous fractional yield of } R$$

$$\Phi = \frac{C_{Rf}}{C_{A0} - C_{Af}} \quad \text{overall yield} = \bar{\varphi} \text{ in the reactor}$$

$$\text{PFR: } \Phi_p = \frac{1}{\Delta C_A} \int_{C_{A0}}^{C_{Af}} \varphi dC_A \quad \text{area under } \varphi(C_A)$$

$$\text{MFR: } \Phi_m = \varphi \text{ evaluated at } C_{Af} \quad \text{one rectangle}$$

$$N \text{ staged MFRs: } \Phi_N = \frac{\sum_i \varphi_i (C_{A,i-1} - C_{A,i})}{C_{A0} - C_{AN}} \quad \text{steps between the two}$$

$$C_{Rf} = \Phi (C_{A0} - C_{Af}) \quad \text{product out of any reactor}$$

Slide 31 draws the three constructions on one φ curve: plug flow shades the area under the curve from C_{Af} to C_{A0} , mixed flow draws the single rectangle of height $\varphi(C_{Af})$ over $(-\Delta C_A)$, and staged reactors climb the curve in steps — the rectangle construction converging to the area as $N \rightarrow \infty$.

6.7 Net rates and the bookkeeping for q reactions

$$r_j = \sum_{i=1}^q r_{ij} \quad \text{net rate of species } j = \text{sum over reactions } i$$

- The deck's example set (slide 34): reaction 1, $A + B \rightarrow 3C + D$; reaction 2, $A + 2C \rightarrow 3E$; reaction 3,

$2B + 3E \rightarrow 4F$; reaction q , $A + \frac{1}{2}C \rightarrow G$. Then $r_A = r_{1A} + r_{2A} + r_{3A} + \dots + r_{qA}$ while r_B picks up only reactions 1 and 3 — species absent from a reaction contribute zero to its sum.

6.8 Worked from the deck

- **Product distribution** (slide 32). $dC_R/dt = 1.0 C_A^{1.5} C_B^{0.3}$, $dC_S/dt = 1.0 C_A^{0.5} C_B^{1.8}$; equal volumetric feeds each at 20 mol/L, 90% conversion of A. The same chemistry through three contacting patterns:

(a) plug flow $C_R = 2.859$ (b) mixed flow $C_R = 4.50$ (c) plug flow, B in side streams $C_R = 7.295$

With equal feeds $C_A = C_B$ everywhere, so $r_R/r_S = C^{-0.5}$: low concentration favours R. Mixed flow then runs entirely at the lowest concentration, plug flow must pass through the high-concentration region making S, and side entrances of B depress C_B relative to C_A further still — hence $2.86 < 4.50 < 7.30$. A factor of 2.6 in product from contacting alone, with no change to chemistry, temperature or throughput.

- **Two MFRs in series** (slide 33), $r_R = k_1 C_A^2$, $r_S = k_2 C_A$, feed $C_{A0} = 1.0$, $C_{S0} = 0.3$, $\tau_1 = 2.5$, $\tau_2 = 10$ min, first-reactor exit $C_{A1} = 0.4$, $C_{R1} = 0.2$, $C_{S1} = 0.7$:

$$C_{A2} = 0.0745, \quad C_{R2} = 0.2277, \quad C_{S2} = 0.9978$$

7 Non-Isothermal Reactor Design

Source: Ch7-_krd.pdf.

7.1 Performance equations and the energy balance

- The isothermal algorithm of section 4 survives unchanged; temperature only adds one unknown, coupled in through Arrhenius. Slide 5 writes the set for a first-order $A \rightarrow B$ PFR:

$$\frac{dX}{dV} = \frac{-r_A}{F_{A0}}, \quad -r_A = k_1 \exp\left[\frac{E}{R}\left(\frac{1}{T_1} - \frac{1}{T}\right)\right] C_{A0}(1-X)$$

- The open-system energy balance (slides 6–8) reads *accumulation = heat added – work done + energy in by flow – energy out by flow*. With $E_i \approx U_i$ (potential and kinetic energies dropped) and flow work absorbed into the enthalpy, steady state gives

$$\dot{Q} - \dot{W}_s + \sum F_{i0} H_{i0} - \sum F_i H_i = 0$$

$$H_i(T) = H_i^\circ(T_R) + C_{p,i}(T - T_R) \quad \text{no phase change, constant } C_p$$

- Substituting the stoichiometry $F_i = F_{A0}(\Theta_i + \nu_i X/a)$ collapses everything to the deck's master equation (slides 9–11):

$$\dot{Q} - \dot{W}_s - F_{A0} X \left[\Delta H_{Rx}^\circ(T_R) + \Delta \hat{C}_p (T - T_R) \right] - F_{A0} \sum_i \Theta_i \tilde{C}_{p,i} (T - T_{i0}) = 0$$

$$\Delta \hat{C}_p = \frac{d}{a} C_{pD} + \frac{c}{a} C_{pC} - \frac{b}{a} C_{pB} - C_{pA}$$

7.2 Adiabatic operation

$$X = \frac{\sum \Theta_i \tilde{C}_{p,i} (T - T_0)}{-\left[\Delta H_{Rx}^\circ(T_R) + \Delta \hat{C}_p (T - T_R) \right]} \quad \text{the energy-balance line } X_{EB}$$

$$T = \frac{X [-\Delta H_{Rx}^\circ(T_R)] + \sum \Theta_i C_{p,i} T_{i0} + X \Delta C_p T_R}{\sum \Theta_i C_{p,i} + X \Delta C_p} \quad \text{solved for } T$$

- The same straight line X_{EB} on the T – X plane holds for **batch**, **CSTR**, **PFR** and **PBR** — the deck brackets all four beside the plot. For an exothermic reaction X_{EB} rises with T ; the reactor can only operate where this line meets what the kinetics and equilibrium allow.

7.3 The equilibrium ceiling, from thermodynamics

Slides 35–45 develop the ceiling rather than asserting it:

$$\begin{aligned}
 dG &= Vdp - SdT + \sum_j \mu_j dn_j, & \mu_j &= \left(\frac{\partial G}{\partial n_j} \right)_{T,P,n_i} \\
 \Delta G &= \Delta G^\circ + RT \ln \left(\prod_j a_j^{\nu_j} \right), & K_a &= \left(\prod_j a_j^{\nu_j} \right)_{eq} = \exp \left(\frac{-\Delta G^\circ}{RT} \right) \\
 K_p &= \left(\prod_j P_j^{\nu_j} \right), \quad K_f = \left(\prod_j f_j^{\nu_j} \right), \quad K_C = \left(\prod_j C_j^{\nu_j} \right) & & \text{activity specialised} \\
 \frac{d \ln K_a(T)}{dT} &= \frac{\Delta H}{RT^2} & & \text{van 't Hoff} \\
 K_p &= \left(\prod_j \left[P \frac{N_{j0} + \nu_j \alpha}{N_{T0} + \alpha \sum_j \nu_j} \right]^{\nu_j} \right) = F(\alpha) & & \text{extent form}
 \end{aligned}$$

- **Reversible exothermic:** X_e falls with T (the equilibrium curve descends); **reversible endothermic:** X_e rises with T . The adiabatic energy-balance line crosses the equilibrium curve; that intersection is the *adiabatic equilibrium conversion*, and no catalyst moves it (slide 46).
- **Worked intersection** (slides 47–48): $A \rightleftharpoons B$, pure A at 300 K, $H_{A0} = -40\,000$, $H_{B0} = -60\,000$ cal/mol, $C_{pA} = C_{pB} = 50$ cal/mol K, $K_C(298) = 10^5$:

$$\begin{aligned}
 X_{EB} &= \frac{C_p(T - T_0)}{-\Delta H_{Rx}} = 2.5 \times 10^{-3}(T - 300) & \left(\frac{50}{20\,000} \right) \\
 K_C(T) &= 100\,000 \exp \left[-33.78 \frac{T - 298}{T} \right] & \left(\frac{|\Delta H|}{R \cdot 298} = \frac{10\,065}{298} = 33.78 \right) \\
 X_e &= \frac{K_C(T)}{1 + K_C(T)} & \text{intersection: deck graphic 0.42 at 465 K}
 \end{aligned}$$

Solving the two printed equations exactly gives $X = 0.401$ at 460.4 K — the deck's 0.42/465 K is a graphical read-off of the same pair, reproduced by `krd-answers-check.py`. The deck's K_C table (slide 47: 661.6 at 350 K down to 0.12 at 500 K) is the same formula evaluated.

7.4 Heat exchange in a tubular reactor

$$\begin{aligned}
 \frac{dT}{dV} &= \frac{Ua(T_a - T) + (-r_A)(-\Delta H_{Rx})}{\sum F_i C_{p,i}} & & \text{no radial gradients} \\
 \text{co-current coolant: } \frac{dT_a}{dV} &= \frac{Ua(T - T_a)}{\dot{m}_c C_{p,c}}
 \end{aligned}$$

- If the coolant flow is large compared with the heat released, T_a is effectively **constant** along the reactor (slide 26); heat added for endothermic, removed for exothermic.
- **Acetic anhydride** (slides 27–30): acetone cracking $\text{CH}_3\text{COCH}_3 \rightarrow \text{CH}_2\text{CO} + \text{CH}_4$, first order with $\ln k = 34.34 - 34\,222/T$, 7850 kg/h through a bank of 1000 1-inch schedule-40 tubes, inlet 1035 K, 162 kPa. Three cases: adiabatic — the endothermic reaction cools itself and *dies after* ≈ 3.5 m³; constant $T_a = 1150$ K with $U = 110$ J/m²s K — T dips then recovers and X climbs; variable heating medium — T_a slides down with T .
- **Ethane cracking at equilibrium** (slide 31): ethane + steam, 2.99 atm, 880 °C, 95 m tubes, 60 % conversion — the deck's specification sheet for the equilibrium-limited endothermic case.

7.5 The CSTR with heat exchange — two solution routes

With $\kappa = UA/(F_{A0}C_{p0})$ and $T_c = (\kappa T_a + T_0)/(1 + \kappa)$:

Specify X $T = T_c + \frac{(-\Delta H_{Rx}^\circ)(X)}{C_{p0}(1 + \kappa)}$, then $k = Ae^{-E/RT}$, $-r_A = kC_{A0}(1 - X)$, $V = F_{A0}X/(-r_A)$

Specify T $X = \frac{C_{p0}(1 + \kappa)(T - T_c)}{-\Delta H_{Rx}^\circ}$, then the same rate-law chain to V

Specify V generate T vs X_{EB} from the energy balance and T vs X_{MB} from $X_{MB} = \tau k/(1 + \tau k)$; the intersection is the operating point

$G(T) = X(-\Delta H_{Rx}) = \frac{\tau Ae^{-E/RT}}{1 + \tau Ae^{-E/RT}} (-\Delta H_{Rx})$	heat generated, first order
$R(T) = C_{p0}(1 + \kappa)(T - T_c)$	heat removed

- Raising T_0 slides the straight $R(T)$ line to the right; raising κ steepens it and moves T_c from T_0 ($\kappa = 0$) toward T_a ($\kappa \rightarrow \infty$) (slides 77–78). Raising τ or E sharpens the S-shape of $G(T)$ (slide 81) — which is what creates multiple intersections.

7.6 Multiple steady states, ignition–extinction, runaway

- Intersections of $G(T)$ and $R(T)$ are the steady states; sweeping T_0 traces the **ignition–extinction curve**: upper and lower branches are attainable, the middle branch is not (slides 82–84).
- **Stability** (slides 85–89): a state is stable where a temperature bump removes more heat than it generates. The deck’s criterion: with

$$\Delta T_{bc} = T' - T_c = \frac{RT_c^2}{E}, \quad T' = \frac{E}{2R} \left[1 - \sqrt{1 - 4 \frac{RT_c}{E}} \right], \quad T_c = \frac{\kappa T_a + T_0}{1 + \kappa},$$

exceeding ΔT_{bc} tips the reactor to the upper state. For industrial reactions $E/RT \approx 16$ –24, so $\Delta T_{bc} \approx 15$ –30 °C.

- With $S' = dG/dT$ at the steady state, **runaway** when $C_{p0}(1 + \kappa) < S'$ (slides 88, 90–91): decreasing UA flattens $R(T)$ until tangency disappears and the CSTR runs away to the upper state. The deck demonstrates it on $2\text{Na}_2\text{S}_2\text{O}_3 + 4\text{H}_2\text{O}_2 \rightarrow \text{Na}_2\text{S}_3\text{O}_6 + \text{Na}_2\text{SO}_4 + 4\text{H}_2\text{O}$ (slides 85–86).
- **Zero vs first order** (slides 92–93): with $\tau = 0.1$ min, $E = 10\,000$ cal/mol, $\Delta H_{Rx} = -10\,000$ cal/mol and inerts 2:1, the deck’s table gives $dE/dT = 80$; the first-order reaction is globally stable for all T_c , while the zero-order one runs away above an entering temperature of 460 K and becomes stable again only past 685 K.

7.7 Interstage cooling, interstage heating, optimum feed temperature

- **Exothermic reversible**: several adiabatic stages with *coolers* between them; each stage starts cooler, further below its equilibrium curve, and the zigzag path on the X – T diagram climbs above what one adiabatic bed can reach (slides 49, 52).
- **Endothermic reversible**: mirror image — *heaters* between stages (slide 50).
- **Optimum feed temperature** (slides 53–54): at low T_0 the rate is slow and conversion is low; at high T_0 the equilibrium conversion collapses and conversion is again low; between them $X(T_0)$ passes through a *maximum* — the optimum feed temperature for a fixed size or catalyst weight.

7.8 Multiple reactions with heat effects (PFR/PBR)

$\frac{dT}{dV} = \frac{Ua(T_a - T) + \sum_i (-r_{ij})(-\Delta H_{Rx,ij})}{\sum_j F_j C_{p,j}}$	coupled with the mole balances and rate laws
--	--

Slide 95’s example: $A \rightarrow B$ ($-r_{1A} = k_{1A}C_A$) and $2A \rightarrow C$ ($-r_{2A} = k_{2A}C_A^2$), pure A at 100 mol/s, 150 °C, 0.1 mol/dm³; $\Delta H_1 = -20\,000$, $\Delta H_2 = -60\,000$ J/mol A; $C_{pA} = C_{pB} = 90$, $C_{pC} = 180$ J/mol °C; $k_{1A} = 10 \exp[(E_1/R)(1/300 - 1/T)]$ s^{−1} with $E_1/R = 4000$ K; $k_{2A} = 0.09 \exp[(E_2/R)(1/300 - 1/T)]$ dm³/mols with $E_2/R = 9000$ K; $Ua = 4000$ J/m³s °C, $T_a = 100$ °C.

7.9 Worked from the deck

- **Butane isomerisation, adiabatic PFR** (slides 14–24). 163 kmol/h of 90 % n-butane/10 % i-pentane at 330 K to 70 %; $\Delta H_{Rx} = -6900$ J/mol, $E/R \approx 7906$ K, $C_{p,\text{butane}} = 141$, $C_{p,\text{pentane}} = 161$ J/mol K. The energy balance gives $T = 330 + 43.3X$ (slope $6900/158.9 \approx 43.3$), and the deck's own explicit set is

$$K_C = 3.03 e^{-830.3(T-333)/(T \cdot 333)}, \quad k = 31.1 e^{7906(T-360)/(T \cdot 360)}$$

$$X_e = \frac{K_C}{1 + K_C}, \quad -r_A = kC_{A0} \left(1 - \left(1 + \frac{1}{K_C}\right)X\right)$$

X	0	0.2	0.4	0.6	0.65	0.7
T (K)	330	338.7	347.3	356.0	358.1	360.3
k (h^{-1})	4.22	7.76	14.02	24.27	27.74	31.67
X_e	0.76	0.74	0.73	0.72	0.718	0.715
$F_{A0}/(-r_A)$ (m^3)	3.74	2.78	2.50	3.88	5.99	23.29

$$V_{X=0.70} \approx 1.97 \text{ m}^3$$

the rate peaks mid-reactor ($-r_A \approx 59$ at $X \approx 0.4$) then collapses toward X_e

- **PFR with co-current heat exchange** (slides 61–67): same chemistry with $Ua = 5000$ kJ/h m^3K , inlet 305 K, coolant 310 K; the deck integrates $dT_a/dV = 5000(T - T_a)/14000$ alongside $dT/dV = (r_A \Delta H - Ua(T - T_a))/(C_{p0}F_{A0})$ — the rate peak is cut and T never exceeds the 325 K limit.
- **Interstage cooling** (slides 48–52), 40 mol/s cooled to 350 K between stages, each stage to 95 % of its equilibrium conversion:

$$\text{R1: } X = 0.397 \text{ at } 458.9 \text{ K; } \quad \text{R2: } X = 0.621 \text{ at } 439.5 \text{ K}$$

$$\dot{Q}_1 = 217.8 \text{ kcal/s, } \quad \dot{Q}_2 = 179.1 \text{ kcal/s}$$

The second stage exits *cooler* than the first (439.5 vs 458.9 K) yet reaches *higher* conversion — the direct payoff of resetting the distance to equilibrium.

- **Propylene oxide, CSTR** (slides 57–74). 43.04 lbmol/h PO into the 300-gal CSTR, first order with

$$k = 16.96 \times 10^{12} e^{-32400/(1.987T)} \text{ h}^{-1}, \quad \tau = 0.1229 \text{ h,}$$

plant ceiling 125 °F:

Case	X	T (R)	T (F)
adiabatic	0.842	610.8	151
with cooling coil	0.364	563.7	104

The adiabatic point sits above the 125 °F = 585 R ceiling — “we can’t use the existing system.” The 40 ft² coil at 85 °F with $U = 100$ Btu/h ft² F pulls the operating point down to the deck’s fsolve values $X = 0.364$, $T = 563.7$ R, inside the constraint. The X_{MB}/X_{EB} table of slide 59 crosses at 613 R, 0.85 — the graphical version of the adiabatic row.

- **Parallel reactions with heat effects** (slide 95). The coupled ODE set of §7.8 integrated to $V = 10^4$ dm³ gives $T \rightarrow 645$ K with the product essentially B.

8 Catalysis and Catalytic Reactors

Source: Catalysis_Krd.pdf.

8.1 What a catalyst does — and what it cannot do

- **Catalyst** (slide 5): a substance that increases the *rate* at which a chemical system approaches equilibrium, *without being substantially consumed*. Selecting one is *not well understood* — in practice, extensive trial and error.

- **Homogeneous** — catalyst in the same phase as the reactants: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \xrightarrow{\text{NO}(\text{g})} 2\text{SO}_3(\text{g})$; sucrose + $\text{H}_2\text{O} \xrightarrow{\text{H}^+(\text{aq})}$ glucose + fructose. **Heterogeneous** — different phase: $4\text{NH}_3 + 5\text{O}_2 \xrightarrow{\text{Pt}, 800^\circ\text{C}} 4\text{NO} + 6\text{H}_2\text{O}$.

The catalyst changes the PACE, not the thermodynamics (slide 15): ΔG and K_{eq} of the reaction are untouched. The deck's example, $\text{CH}_4(\text{g}) + \text{CO}_2(\text{g}) = 2\text{CO}(\text{g}) + 2\text{H}_2(\text{g})$:

$\Delta G_{100^\circ\text{C}} = +151 \text{ kJ/mol}$	no driving force: no reaction, with or without catalyst
$\Delta G_{700^\circ\text{C}} = -16 \text{ kJ/mol}$	driving force present, but the mixture still does not react on its own

At 700°C with Pt/ZrO₂ or Ni/Al₂O₃ in the reactor, equilibrium conversion ($\sim 100\%$) is reached at that same temperature.

8.2 Solid catalysts: active phase, promoters, supports

- A solid catalyst is built from three roles (slide 16): the **active phase** where reaction occurs (mostly metal/metal oxide); a **promoter** — textual (e.g. Al for NH₃ production), electric/structural, or poison-resistant; and a **support/carrier** giving mechanical strength and surface area (98% of the surface is inside the porous structure; the support may or may not be catalytically active). **Gauze/foil**: external active sites alone are enough for very fast rates.
- **Support materials** (slide 17): alumina — inexpensive, 1–700 m²/g, acidic/neutral; silica — 100–800 m²/g, acidic; **zeolite** — mixture of alumina and silica, often exchanged metal ion present, *shape selective*, acidic; others: active carbon (up to 1000 m²/g), titania (10–50), zirconia (10–100), magnesia (10), lanthana (10).
- **Specificity** (slides 21, 28): the choice of catalyst selects the product. Ethanol over copper dehydrogenates, $\text{C}_2\text{H}_5\text{OH} \rightarrow \text{CH}_3\text{CHO} + \text{H}_2$; over alumina it dehydrates, $\text{C}_2\text{H}_5\text{OH} \rightarrow \text{C}_2\text{H}_4 + \text{H}_2\text{O}$. The distance between active sites and their match to bond lengths decides which bonds are weakened.

8.3 The seven steps; the three intrinsic stages

External diffusion → internal diffusion → adsorption of A → surface reaction → desorption of B → internal diffusion out → external diffusion out. Assume one step limits, write its rate, and combine with the **site balance**

$$C_t = C_v + \sum_i C_{i \cdot S}$$

Then test the assumption: if the assumed rate-limiting step does not reproduce the data, move to the next one. The deck works through each assumption in turn.

- Inside the film and the pore the three intrinsic stages are **adsorption** → **reaction** → **desorption** (slides 24–25): adsorption lands the species on an active site and weakens its bonds; the surface holds adsorbed gases in the right orientation so favourable collisions become likely; desorption releases the products after an electron rearrangement.

	Physisorption	Chemisorption
force	van der Waals	chemical bond
adsorbed layers	multilayer	only one layer
heat of adsorption	low (10–40 kJ/mol)	high (> 40 kJ/mol)
selectivity	low	high
temperature at which it occurs	low	high

Slide 27. Chemisorption is the catalytically useful one: a real bond to the surface is what activates the molecule.

8.4 Mechanisms: Langmuir–Hinshelwood and Eley–Rideal

- **Langmuir–Hinshelwood** (slides 29–30): adsorption from the gas phase, reactions *between adsorbed molecules*, dissociation at the surface, desorption. For $\text{A}_2 + 2\text{B} = 2\text{AB}$ the deck writes



- **Eley–Rideal** (slides 31–32): the same adsorption steps, but the last step is the direct reaction between the adsorbed molecule A^* and a *gas-phase* molecule B : $A^* + B = AB + *$. That fourth step cannot occur in an L–H mechanism.
- **Telling them apart** (slide 33), rate versus coverage $\theta_{A,S}$: for E–R the rate *increases* with coverage until the surface is completely covered by A^* ; for L–H the rate goes through a *maximum* and ends at zero when A^* blocks every site — because the step $B + * = B^*$ requires a free site. Whether a real reaction follows one or the other cannot be decided without experimental or computational studies.

Intrinsic-kinetics algorithm (slide 34): write a rate equation for every step; assume one step is the RDS — the intrinsically slowest; set $r_o = r_{\text{RDS}}$; take all other steps at equilibrium ($r \approx 0$); express every surface-adsorbed species concentration in terms of bulk species; write the site balance; substitute the adsorbed-species equations and C_v into r_{RDS} .

- **Ammonia synthesis mechanism as printed** (slide 36): $\text{H}_2 + 2S \rightarrow 2\text{H}\cdot\text{S}$ (rapid); $\text{N}_2 + S \rightleftharpoons \text{N}_2\cdot\text{S}$ and $\text{N}_2\cdot\text{S} + S \rightarrow 2\text{N}\cdot\text{S}$ (**rate-limiting**); then $\text{N}\cdot\text{S} + \text{H}\cdot\text{S} \rightleftharpoons \text{HN}\cdot\text{S} + S$, $\text{NH}\cdot\text{S} + \text{H}\cdot\text{S} \rightleftharpoons \text{H}_2\text{N}\cdot\text{S} + S$, $\text{H}_2\text{N}\cdot\text{S} + \text{H}\cdot\text{S} \rightleftharpoons \text{NH}_3\cdot\text{S} + S$, $\text{NH}_3\cdot\text{S} \rightleftharpoons \text{NH}_3 + S$ (rapid).
- **Cyclohexane dehydrogenation** (slide 35): $\text{C}_6\text{H}_{12} \xrightarrow[\text{Al}_2\text{O}_3]{\text{Ni}} \text{C}_6\text{H}_6 + 3\text{H}_2$. The measured rate rises with concentration and saturates — the signature of an adsorption denominator — and the task is to (1) propose the mechanism and (2) find the RDS consistent with the data.

8.5 Adsorption isotherms and rate forms

$$C_{A\cdot S} = \frac{K_A P_A C_t}{1 + K_A P_A + K_B P_B} \quad \text{Langmuir, competitive sites}$$

$$-r'_T = \frac{k P_T P_{\text{H}_2}}{1 + K_B P_B + K_T P_T} \quad \text{Eley–Rideal type}$$

- **Langmuir–Hinshelwood** assumes both reactants are adsorbed before reacting; **Eley–Rideal** assumes one reacts directly from the gas phase. The denominator is the site balance in both cases — that is what makes the rate saturate at high pressure.

8.6 Pore structure and the effective diffusivity

- **Pore classes** (slide 38): mesopores < 2 nm; micropores 2 to 50 nm; macro pores > 50 nm; the active component coats the pore walls. Pores are of random size except in zeolites.

The deck's micro/meso labels are inverted relative to IUPAC (IUPAC: micro < 2 nm, meso 2–50 nm). Reported as printed, not silently corrected — the sizes, not the names, are what the calculations use.

$$D_e = \frac{D_{AB} \phi_p \sigma_c}{\tilde{\tau}} \quad \text{effective diffusivity}$$

$$\tilde{\tau} = \frac{\text{actual distance travelled}}{\text{shortest distance}} \quad \text{tortuosity; } \beta = \frac{\text{area } A_2}{\text{area } A_1}, \quad \sigma_c = f(\beta)$$

$$90^\circ \text{ bent pore: } \tilde{\tau} = \frac{2L}{\sqrt{2}L^2} = \sqrt{2} = 1.414 \quad \text{worked on slide 40}$$

8.7 Internal diffusion: Thiele modulus, effectiveness factor, geometries

$$\frac{d^2\psi}{d\lambda^2} + \frac{2}{\lambda} \frac{d\psi}{d\lambda} - \phi_1^2 \psi = 0 \quad \psi = \frac{C_A}{C_{As}}, \quad \lambda = \frac{r}{R}, \quad \phi_1^2 = \frac{k_1 R^2 C_{As}^{n-1}}{D_e} \quad (\text{slide 42})$$

$$\phi_s = R \sqrt{\frac{k_1}{D_e}} \quad \text{Thiele modulus, first order, sphere}$$

$$\eta = \frac{3}{\phi^2} (\phi \coth \phi - 1) \quad \text{effectiveness factor, sphere}$$

$$\eta = \frac{\tanh \phi}{\phi} \quad \text{slab geometry}$$

$$\frac{C_A}{C_{As}} = \frac{R}{r} \cdot \frac{\sinh(\phi r/R)}{\sinh \phi} \quad \text{concentration profile inside the pellet}$$

- $\eta \rightarrow 1$ for small ϕ (reaction-limited: the whole pellet is used); $\eta \rightarrow 3/\phi$ for large ϕ (diffusion-limited: only a shell near the surface reacts).
- **Other geometries** (slide 44), with $\omega = x/x_p$ or $\bar{r}/R_p$ and $\phi = x_p$ or R_p times $\sqrt{k/D_{T,A}^e}$:

Slab	Cylinder	Sphere
$\psi \quad \frac{\cosh(\phi\omega)}{\cosh \phi}$	$\frac{I_0(\phi\omega)}{I_0(\phi)}$	$\frac{\sinh(\phi\omega)}{\omega \sinh \phi}$
$\eta \quad \frac{\tanh \phi}{\phi}$	$\frac{2 I_1(\phi)}{\phi I_0(\phi)}$	$\frac{3}{\phi} \left[\frac{1}{\tanh \phi} - \frac{1}{\phi} \right]$

I_i is the modified Bessel function of order i .

- **Non-isothermal pellet** (slide 46): with $\gamma = E/(RT_s)$ (Arrhenius number) and $\beta = \Delta T_{max}/T_s = -\Delta H_{Rx} D_e C_{As} / (k_s T_s)$, an exothermic reaction can run hotter inside the pellet than at its surface, and η can exceed 1.

8.8 Falsified kinetics and the diagnostic criteria

- **Falsified kinetics** (slide 45): cumene cracking on $\text{SiO}_2\text{-Al}_2\text{O}_3$, measured rate against $1/T$ for four pellet radii. Small pellets give the steep, true-activation-energy line; as the pellet grows the line drops and its slope falls toward $E/2$ — the Arrhenius plot itself tells you when internal diffusion has taken over.
- **Weisz–Prater** (slide 49): evaluate limitation from the *observed* rate alone,

$$C_{WP} = \eta \phi^2 = \frac{(-r'_A)_{\text{obs}} \rho_c R^2}{D_e C_{As}} = \frac{\text{actual reaction rate}}{\text{a diffusion rate}}$$

$$C_{WP} \ll 1$$

no concentration gradients in the pellet

$$C_{WP} \gg 1$$

internal diffusion controls the reaction

- **Mears criteria** (slide 56) for the *external* film:

$$\frac{(-r'_A) \rho_b R n}{k_c C_{Ab}} < 0.15$$

external concentration gradients negligible

$$\frac{|-\Delta H_{Rx} (-r'_A) \rho_b R E|}{h T^2 R_g} < 0.15$$

external temperature gradients negligible

n = order of reaction, R = pellet radius, ρ_b = bulk density, h = heat-transfer coefficient between gas and pellet. Weisz–Prater is the practical internal test because it needs only the observed rate; Mears is its external counterpart.

8.9 External mass transfer and the overall effectiveness factor

- **Mass-transfer coefficient** (slide 51): $k_c = 0.6 \frac{D_{AB}^{2/3}}{\nu^{1/6}} \times \frac{U^{1/2}}{d_p^{1/2}}$. Plotting $-r'_A$ against $(U/d_p)^{1/2}$: the rate first rises with velocity (**diffusion limited**, the boundary layer thins) and then plateaus (**reaction limited**, the surface reaction is the bottleneck).
- **Steady-state matching** (slides 53–54): the molar rate of transfer from the bulk to the external surface equals the net rate of reaction on and within the pellet, $M_{Ar} = -r'_A$ (external area + internal area), with external area = $a_c \Delta V$ and internal area = $S_a \rho_b \Delta V$.

$$\Omega = \frac{\text{actual overall rate}}{\text{rate if the whole surface saw } C_{Ab}, T_b} = \frac{\eta}{1 + \eta k_1'' S_a \rho_b / (k_c a_c)}$$

8.10 Packed beds: design with diffusion and reaction

- The packed bed is the workhorse (slide 58) — steam reforming, catalytic reforming, isomerization, polymerization, ammonia and SO_3 synthesis — run adiabatic or non-adiabatic, particle sizes $1/8''$ to $1/4''$.
- **Balance over a bed slice** (slide 60): $A_c W_{Az}|_z - A_c W_{Az}|_{z+\Delta z} + r'_A \rho_b A_c \Delta z = 0$, i.e. $-\frac{dW_{Az}}{dz} + r'_A \rho_b = 0$. With a dispersion term this becomes

$$D_a \frac{d^2 C_{Ab}}{dz^2} - U \frac{dC_{Ab}}{dz} - \Omega \rho_b k'' S_a C_{Ab} = 0$$

$$X = 1 - \frac{C_{Ab}}{C_{Ab0}} = 1 - e^{-(\rho_b k'' S_a \Omega L)/U} \quad \text{dispersion neglected}$$

- **Configurations** (slide 59): **multi-bed** (unit operations between PBRs); **reverse flow** — for exothermic reactions the temperature rise along the bed can deactivate catalyst, so the flow direction is switched periodically, which also preheats the feed; **counter-flow** — two chambers, a lean stream preheats the fluid and raises the rate; **radial flow** — feed enters and flows radially through an annulus of catalyst, for operation under low pressure drop.

8.11 Worked from the deck

- **Concentration inside a pellet** (slide 48), sphere $d_p = 2 \times 10^{-3}$ cm, $D_e = 0.1$ cm²/s:

$$\phi = 5.986, \quad \frac{C_A}{C_{As}} = 0.2371, \quad C_A = 2.37 \times 10^{-4}$$

$$\text{to reach } C_A/C_{As} = 0.8 : \quad \phi = 2.042 \quad \Rightarrow \quad d_{\text{new}} = 6.82 \times 10^{-4} \text{ cm}$$

At $\phi = 5.99$ only 24 % of the surface concentration survives at the pellet centre. Shrinking the diameter to about a third of the original is what it takes to make the whole pellet effective.

- **Apparent activation energy** (slide 57), $R = 0.2$ cm, $D_e = 0.015$ cm²/s, $100 \rightarrow 150$ °C:

T	ϕ	η	r
100 °C	1.575	0.8660	2.617×10^{-5}
150 °C	7.75	0.337	2.468×10^{-4}

$$E_{\text{app}} = 14080 \text{ cal/mol}$$

The measured activation energy is roughly *half* the true one because η is falling as fast as k is rising — the classic signature of internal diffusion control, the falsified kinetics of slide 45 in numbers. Halving the observed E is the diagnostic, not an error in the Arrhenius plot.

9 Non-Ideal Reactors and Residence-Time Distributions

Source: Non_ideal_KRD.pdf — headed *Reaction Engineering / CHE G641*, the graduate treatment of this material.

9.1 Where ideality breaks

- **Why it matters (slides 3–4)**: the ideal design equations assume identical residence times (PFR/PBR, no axial diffusion) or perfect mixing (CSTR/batch); real vessels sit in between — non-uniform flow, inefficient or absent mixing, bypassing and dead zones change conversion, yield and the temperature development of the process.
- **CSTR (slide 5)**: the feed entrance runs at relatively high concentration while stagnant *dead zones* run low; part of the flow can short-circuit the vessel as bypass.
- **PBR (slide 6)**: ideal plug flow moves every molecule at the same axial rate; the real bed adds turbulent velocity profiles, back-mixing and molecular diffusion, so molecules travel at different speeds and in different directions (channeling, slide 14).

- **Micro vs macro mixing (slides 8–9):** macromixing is the RTD itself; micromixing asks whether molecules of different ages meet at all — *complete segregation*, same-age globules leaving together, versus *complete micromixing* — and whether mixing occurs early or late along the flow path.

9.2 The RTD functions

$E(t) = \frac{C(t)}{\int_0^\infty C dt}$	normalised tracer response
$F(t) = \int_0^t E dt$	cumulative fraction younger than t
$t_m = \int_0^\infty t E dt = \frac{V}{v_0}$	mean residence time
$\sigma^2 = \int_0^\infty (t - t_m)^2 E dt$	variance, i.e. the spread
$1 - F(t) = \frac{V}{v} I(t)$	internal age distribution $I(t)$

Normalising by the area is what makes $E(t)$ a probability density, and it is the step most often missed when a pulse experiment is worked up by hand.

The profiles are a diagnostic (slides 14, 21): a sharp spike reads as a nearly ideal PFR, a decaying exponential as a nearly ideal CSTR, an early spike with a long tail as a PBR with channeling and dead zones, and a two-humped curve as a CSTR with dead zones.

9.3 Ideal limits

PFR: $E(t) = \delta(t - \tau)$	all fluid elements the same age
CSTR: $E(t) = \frac{e^{-t/\tau}}{\tau}$	exponential; some fluid leaves almost at once

9.4 Zero-parameter models: segregation and maximum mixedness

$\bar{X} = \int_0^\infty X_{\text{batch}}(t) E(t) dt$	each fluid element is a small batch reactor
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- $n > 1$: $\bar{X}$ exceeds the ideal-mixed conversion. $0 < n < 1$: the opposite. Complete segregation is therefore a bound, and it tells you which way a real reactor departs from ideal mixing without any further modelling.

The deck's model menu (slide 22): *zero* adjustable parameters — the segregation model above and the maximum-mixedness model, its micromixed opposite extreme; *one* parameter — tanks-in-series and dispersion; *two* parameters — the real vessel as a combination of ideal reactors (§9.7).

9.5 Tanks-in-series model

$E(t) = \frac{t^{n-1}}{(n-1)! \tau_i^n} e^{-t/\tau_i}$	gamma distribution
$n = \frac{t_m^2}{\sigma^2}$	number of tanks, from the tracer data alone
$X = 1 - (1 + k\tau_i)^{-n}$	first-order conversion

Where the gamma form comes from (slides 26–28): a pulse through N equal CSTRs gives $C_1 = C_0 e^{-t/\tau_i}$, $C_2 = C_0 t e^{-t/\tau_i} / \tau_i$, $C_3 = C_0 t^2 e^{-t/\tau_i} / 2\tau_i^2$, and in general $E(t) = t^{N-1} e^{-t/\tau_i} / [(N-1)! \tau_i^N]$, or dimensionless $E(\theta) = N(N\theta)^{N-1} e^{-N\theta} / (N-1)!$. As $N \rightarrow \infty$ the curve collapses to the plug-flow spike, which is why $n = t_m^2 / \sigma^2$ measures the departure from plug flow.

9.6 Dispersion model

$$\begin{aligned}
X &= 1 - \frac{4q e^{Pe/2}}{(1+q)^2 e^{Pe q/2} - (1-q)^2 e^{-Pe q/2}} && \text{closed vessel, first order} \\
q &= \sqrt{1 + \frac{4Da}{Pe}}, && Da = k\tau, \quad Pe = \frac{UL}{D_a} \\
\sigma_\theta^2 &= \frac{2}{Pe} - \frac{2}{Pe^2} (1 - e^{-Pe}) && \text{link from tracer data to } Pe \\
n &= \frac{Pe}{2} + 1 && \text{equivalent number of tanks}
\end{aligned}$$

The picture behind it (slides 29–33): molecules diffuse ahead of and lag behind the molar average velocity, i.e. a Fickian dispersion $D_a dC/dz$ superimposed on the bulk flow, so the tracer pulse broadens as it travels; the mole balance over a slice ΔZ with $F_A = uAC_A + AD_a dC_A/dz$ yields the closed-vessel solution above. The single parameter $Da/(UL)$ is read from a pulse tracer test; the deck's E -curves for $Da/(UL) = 0.002, 0.025, 0.2$ show small, intermediate and large deviation from plug flow.

9.7 Two-parameter models

- A real CSTR is represented as an **ideal reacting region** of volume $V_s = \alpha V$ with a **dead zone** $V_d = (1 - \alpha)V$, plus a **bypass** stream; $\beta = v_s/v_0$ is the fraction of the feed entering the reacting region (slides 40–41). The modelled conversion is

$$1 - X = \beta + \frac{(1 - \beta)^2}{1 - \beta + \alpha\tau k}.$$

- Tracer response takes the form

$$\frac{C_T}{C_{T0}} = 1 - (1 - \beta) e^{-(1-\beta)t/(\alpha\tau)}$$

and linearising it (slide 42),

$$\ln \frac{C_{T0}}{C_{T0} - C_T} = \ln \frac{1}{1 - \beta} + \frac{1 - \beta}{\alpha} \cdot \frac{t}{\tau},$$

so the intercept and slope of the tracer straight line give (α, β) .

- The deck's second two-parameter variant (slides 43–44) models a real CSTR as **two ideal CSTRs with interchange** — a highly agitated region near the impeller exchanging with a quieter one, $V_d = (1 - \alpha)V$ — with

$$X = \frac{(\beta + \alpha\tau k)(\beta + (1 - \alpha)\tau k) - \beta^2}{(1 + \beta + \alpha\tau k)(\beta + (1 - \alpha)\tau k) - \beta^2},$$

and (α, β) again fixed by a tracer test, the response now a sum of two exponentials with exponents m_1, m_2 .

9.8 Worked from the deck

- Pulse tracer** (slide 17):

$$\text{area} = 50.03, \quad t_m = 5.19 \text{ min}, \quad \sigma^2 = 6.03 \quad \implies \quad \boxed{\text{fraction between 3 and 6 min} = 0.513}$$

Just over half the fluid leaves within a minute either side of the mean — the spread is the whole story here, not the mean.

- Dispersion example** (slides 52–59):

$$Pe = 7.74, \quad Da = 1.28$$

Model	X
dispersion (closed vessel)	0.679
tanks-in-series, $n = 4.45$	0.676
ideal PFR	0.720
ideal CSTR	0.560

The two real-reactor models agree to 0.003 and sit *between* the ideal limits, as they must. Using $n = Pe/2 + 1 = 4.87$ instead of the moment-derived 4.45 changes the answer negligibly — but the agreement of the two independent models is what justifies trusting either. Treating this vessel as an ideal CSTR would understate conversion by 0.12; as an ideal PFR, overstate it by 0.04.

$$\begin{aligned}
 X &= \frac{N_{A0} - N_A}{N_{A0}} & \tau &= \frac{V}{v_0} \\
 V_{\text{CSTR}} &= \frac{F_{A0}X}{(-r_A)_{\text{exit}}} & V_{\text{PFR}} &= F_{A0} \int_0^X \frac{dX}{-r_A} \\
 W_{\text{PBR}} &= F_{A0} \int_0^X \frac{dX}{-r'_A} & t_{\text{batch}} &= N_{A0} \int_0^X \frac{dX}{-r_A V} \\
 -r_A &= k C_A^\alpha C_B^\beta & k &= A e^{-E/RT} \\
 C_i &= C_{A0} \frac{\Theta_i + \nu_i X/a}{1 + \varepsilon X} \cdot \frac{T_0}{T} \cdot \frac{P}{P_0} & \varepsilon &= y_{A0} \delta \\
 X_{\text{CSTR,1st}} &= \frac{k\tau}{1 + k\tau} & X_{N \text{ CSTRs}} &= 1 - \frac{1}{(1 + k\tau_i)^N} \\
 t_{1/2} &= \frac{2^{n-1} - 1}{(n-1)k} C_{A0}^{1-n} & \ln t_{1/2} &= (1-n) \ln C_{A0} + \text{const} \\
 S_{D/U} &= \frac{r_D}{r_U} & \tilde{Y}_D &= \frac{F_D}{F_{A0} - F_A} \\
 T &= T_0 + (\text{slope})X & X_e &= \frac{K_C}{1 + K_C} \\
 \phi &= R\sqrt{k_1/D_e} & \eta &= \frac{3}{\phi^2}(\phi \coth \phi - 1) \\
 C_{WP} &= \frac{(-r'_A)_{\text{obs}} R^2}{D_e C_{As}} & n &= \frac{t_m^2}{\sigma^2} = \frac{Pe}{2} + 1
 \end{aligned}$$

The four comparisons worth knowing cold

Question	Better	From
Same conversion, one vessel: PFR or CSTR?	PFR	§2.6 (2.165 vs 6.40 m ³)
Same total volume, two CSTRs in series or parallel?	series	§4.9 (0.900 vs 0.813)
PBR then CSTR, or CSTR then PBR?	CSTR first	§4.13 (0.756 vs 0.629)
Real vessel, or an ideal limit?	neither	§9.8 (0.679 vs 0.720 / 0.560)

Prepared from the nine CHE F311 slide decks named in the box on page 1; every formula above is the form taught in the deck heading its section, and every number is reproduced by `krd-answers-check.py`, which exits 0. Structure and depth follow the course standard set by `SP2-notes.tex` (CHE F313). Five places where the source itself is inconsistent are reported rather than corrected: the pressure-drop figures in §4.14, the fit annotations in §4.8 (both Ch4), the graduate (CHE G641) heading on `Non_ideal_KRD.pdf`, and two Ch6 sentences — the concentration verb on slide 18 (§6.3) and the Case-IV opening on slide 25 (§6.4). Symbols: X conversion, $-r_A$ rate of disappearance of A, $-r'_A$ rate per catalyst mass, F_{A0} inlet molar flow of A, C_A concentration, Θ_i feed ratio C_{i0}/C_{A0} , ε volume-change factor, δ mole-change number, V volume, W catalyst mass, τ space time, Da Damköhler number, k rate constant, A pre-exponential factor, E activation energy, K_C equilibrium constant, X_e equilibrium conversion, ΔH_{Rx} heat of reaction, Ua heat-transfer coefficient times area, T_a coolant temperature, p reduced pressure P/P_0 , α pressure-drop parameter, ϕ bed porosity or Thiele modulus as the section states, η effectiveness factor, D_e effective diffusivity, C_{WP} Weisz–Prater number, $E(t)$ residence-time distribution, $F(t)$ cumulative RTD, t_m mean residence time, σ^2 variance, Pe Péclet number, n tanks-in-series number, $S_{D/U}$ selectivity, $\tilde{Y}_D$ yield.